



Causal Inference in Observational Studies

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Disclaimers: (i) Sections and lines in brown are under construction. (ii) For all mathematical simplifications featuring “= ... =”, the detailed steps are provided in the Appendix.

1 Definitions

A research design is the overall plan, given a research question, for collecting and analyzing data in order to draw credible conclusions.

Let us set the scene for the present document. We are concerned with the branch of econometrics that seeks to identify *causal* effects, particularly from observational data. This is a real challenge: relationships between observed variables are easy to estimate, but how do we know whether they are causal or spurious?

Precisely, we want to estimate the causal effect of an intervention or *treatment* for some population. We need a research design that is able to credibly *identify* the effect. A central component of the research design of causal inference studies is hence the *identification strategy*.¹

An identification strategy is the argument of how observational data are used to recover the causal effect of interest. It specifies notably:

- a source of identifying variation in the treatment variable,
- the assumptions under which this variation identifies the causal effect.

An estimation strategy then specifies the econometric methods used to estimate the identified effect.

Different experiments will provide different types of variation in the treatment. For causal inference, when internal validity is the priority,² a research design based on a randomized trial is commonly regarded as the gold standard. By randomly assigning treatment across units, a randomized trial prevents systematic selection into treatment and makes the treated and the untreated groups comparable in expectation, thereby allowing treatment effects to be identified under the relevant assumptions—this will be detailed in later sections. As observational studies strive to achieve this strength of evidence, their identification strategies generally seek to approximate such an experiment.

Experiments

A true experiment is a study in which the researcher actively assigns or manipulates the treatment and observes its effect on an outcome. In a **randomized experiment**, treatment is randomly assigned, making treatment status independent of potential outcomes in expectation.

A natural experiment is an observational study in which plausibly exogenous^a—often *as-if random*—variation in a treatment D or an instrument Z has occurred—mimicking the exogeneity of a randomized experiment. Researchers don’t create natural experiments, they find them.

Ex: Certain weather events, natural disasters, lotteries, or arbitrary assignment rules.

A quasi-experiment^b is a study of intentional treatment that resembles a randomized experiment but lacks full researcher-controlled random assignment. Because the treated group and the untreated group (often called *comparison* group rather than *control*) may differ systematically, causal identification relies on the study’s comparison structure, design features, and specific identifying assumptions.

Ex: In the 1990s, the U.S. Department of Housing and Urban Development (HUD) implemented a grant program to encourage resident management of low-income public housing projects. Housing projects were *selected* in 11 cities nationwide, so the treatment (the award of HUD funding) was not randomly assigned. But similar housing projects in the same cities provided a reasonably valid comparison, so the HUD was able to evaluate the program.

^aExogeneity is w.r.t. potential outcomes, i.e., the variation is unrelated to the factors that would otherwise determine the outcome.

^b[△] Terminology varies across disciplines; econometricians often use the term ‘quasi-experimental’ to refer to any credible nonrandomized design—including natural experiments.

¹Angrist and Pischke (2008) uses the notions of research design and identification strategy interchangeably.

²The notions of internal vs. external validity are defined in section 4.4.3. Let us note for the time being that the methods of choice for internal validity may actually limit external validity. For example, a zoo is a controllable setting amenable to drawing causal inferences about animal behavior, but those inferences may not generalize to animals in the wild.

2 Framework: a counterfactual approach to causality

2.1 The potential outcomes framework

2.1.1 The original selection bias problem and independence assumptions

Consider a population, of which we observe a sample of units $i = 1, \dots, n$, and a binary treatment $D_i \in \{0, 1\}$ whose causal effect on an outcome Y we want to estimate. Let Y_i^1 denote unit i 's potential outcome³ if they were to receive the treatment, Y_i^0 otherwise, and Y_i their actual realized outcome. We assume that there is no interference between units.⁴ The potential outcomes framework allows for defining estimands:

Individual treatment effects (TEs)	$Y_i^1 - Y_i^0, \forall i$	<i>what we would ideally estimate</i>
Average treatment effect (ATE)	$\mathbb{E}[Y_i^1 - Y_i^0]$	<i>what we reasonably want to estimate</i>
Average treatment effect on the treated (ATET)	$\mathbb{E}[Y_i^1 - Y_i^0 D_i=1]$	"
Conditional average treatment effect (CATE)	$\mathbb{E}[Y_i^1 - Y_i^0 X_i]$	"
Difference in mean observed outcomes	$\mathbb{E}[Y_i D_i=1] - \mathbb{E}[Y_i D_i=0]$	<i>what we can estimate</i>
----- <i>All expectations can be defined conditional on covariates X, yielding then conditional means within strata.</i>		

The original selection bias problem To measure $TE_i = Y_i^1 - Y_i^0$, we need to observe the same individual with and without treatment. This is impossible, we can never observe the counterfactual.⁵ What we can only estimate is the difference in means of observed outcomes, but this quantity contains a selection bias, stemming from the difference in mean Y_i^0 between the treated and the untreated:⁶

$$\mathbb{E}[Y_i | D_i=1] - \mathbb{E}[Y_i | D_i=0] = \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 | D_i=1]}_{\text{ATET}} + \underbrace{\mathbb{E}[Y_i^0 | D_i=1] - \mathbb{E}[Y_i^0 | D_i=0]}_{\text{selection bias}}$$

Identification, i.e., elimination of the selection bias, can be achieved through analysis, depending on the *treatment assignment* mechanism.⁷ We distinguish three classes of assumed assignment mechanisms:

- *Random assignment*

If treatment is randomly assigned, it is independent of potential outcomes: $D_i \perp\!\!\!\perp (Y_i^0, Y_i^1)$, so there is no selection bias *in expectation*.⁸ Since independence also implies $ATET = ATE$, this independence assumption (IA) identifies the ATE.

- *Selection on observables*

Generally, in observational studies, $D_i \not\perp\!\!\!\perp (Y_i^0, Y_i^1)$. However, if we can *match* treated and untreated units to be proper counterfactuals, i.e., if conditional on some pre-treatment characteristics X_i , treatment assignment is independent of potential outcomes: $D_i \perp\!\!\!\perp (Y_i^0, Y_i^1) | X_i$, and if there are both treated and control units across the support of X_i (i.e., overlap), then we can again eliminate selection bias in expectation. We compare outcomes *for given* X_i , i.e., within stratum:

$$\underbrace{\mathbb{E}[Y_i | D_i=1, X_i] - \mathbb{E}[Y_i | D_i=0, X_i]}_{\text{difference in mean outcomes given } X_i} = \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 | D_i=1, X_i]}_{\text{ATET given } X_i: \text{CATE}} + \underbrace{\mathbb{E}[Y_i^0 | D_i=1, X_i] - \mathbb{E}[Y_i^0 | D_i=0, X_i]}_{\text{selection bias given } X_i}$$

³The potential outcomes framework for causal inference builds on [Neyman \(1923\)](#), was extended to observational studies by [Rubin \(1974\)](#), and became popular in econometrics around 1990.

⁴The Stable Unit Treatment Value Assumption (SUTVA) states that the treatment received by one unit does not affect the potential outcomes of other units. This notably excludes spillovers and strategic interactions. When this is unrealistic, alternative approaches include redefining the unit of interest (aggregating), or directly modeling the interactions.

⁵This is the *fundamental problem of causal inference*. Its implication: we never observe causal effects.

⁶The selection bias is the sum of the net average balance (i.e., the net difference of means) of other causes across the two groups. For example, if individuals with low Y_i^0 choose treatment more frequently, then $\mathbb{E}[Y_i^0 | D_i=1] < \mathbb{E}[Y_i^0 | D_i=0]$; comparing Y between treated and untreated underestimates the TE. Say we look at the effect of hospitalization; sick individuals go to the hospital (get treated) more often than healthy individuals. But they would also have been less healthy had they stayed at home.

⁷[Imbens and Wooldridge \(2009\)](#) formally defines the assignment mechanism as $\Pr(D_i=1 | Y_i^0, Y_i^1, X_i)$.

⁸Independence removes selection bias *in this expectation form*, where the expectation is taken over repeated randomizations on the trial sample, each with its own allocation of treatments and controls ([Deaton and Cartwright, 2018](#)). Independence does not imply actual balance in any single trial. Therefore in any RCT, the sample analog of the last term may not actually be 0.

The conditional independence assumption (CIA) eliminates selection bias, and also implies CATET = CATE, thereby identifying the CATEs. We may then combine these CATEs by weighting them in some way to recover a single weighted average.⁹

- *Selection on unobservables*

We can't assume that we observe all factors that are associated with both the assignment and the potential outcomes, i.e., the CIA fails. To identify some treatment effect(s), we need other identification strategies whose design-specific assumptions will identify design-specific estimands, often more local than the ATE. The most common methods are described in section 3.

The credible identification of treatment effects requires **identifying assumptions** that make a counterfactual comparison valid.

- Independence assumptions¹⁰ are the strongest:
 - The IA identifies the ATE.
 - The {CIA + overlap} identify CATEs—and from there, chosen aggregates.
- Alternative assumptions may also yield identification:
 - *Instrumental variables* – {An exogenous relevant instrument Z for D + exclusion and monotonicity} identify the LATE for compliers.
 - *Regression discontinuity* – Continuity of the potential-outcome conditional means at a treatment threshold identifies a (L)ATE at the threshold.
 - *Difference-in-differences* – {Parallel counterfactual trends + no anticipation} identify an ATET for a specified group and period.

Beyond average effects: distributional effects The identification achieved through independence assumptions extends beyond average treatment effects. Independence implies that the full marginal distribution of each potential outcome is identified. We can hence identify quantile treatment effects (QTEs), which are the effects of the treatment at corresponding quantiles of the two marginal potential-outcome distributions: $\tau_p := Q_{Y^1}(p) - Q_{Y^0}(p)$, $\forall p \in [0, 1]$.¹¹ QTEs can reveal distributional treatment effects obscured by the ATE, e.g., effects concentrated in the tails of the outcome distribution.

Note that under the CIA, one identifies *conditional* quantile treatment effects, $\tau_p(x) := Q_{Y^1|X=x}(p) - Q_{Y^0|X=x}(p)$. However, in general, the unconditional QTE $\tau_p \neq \mathbb{E}_X[\tau_p(X)]$. Consequently, τ_p cannot generally be obtained by averaging conditional QTEs. One must instead average the conditional distribution functions over X and then invert the resulting marginal distribution functions (Imbens and Wooldridge, 2009).

2.1.2 Average treatment effects as regression slopes and endogeneity

Let $\beta_i := Y_i^1 - Y_i^0$ denote the individual treatment effect for unit i . We distinguish a causal regression-form decomposition from the population OLS projection:

- *Causal regression-form decomposition:* The relation between observed and potential outcomes can be

⁹For instance, the matching estimand will weight them by the distribution of X among the treated, which recovers the ATET, whereas the linear regression estimand will weight them by the conditional variance of D , emphasizing strata with substantial treatment-control overlap — see section 2.1.2.

¹⁰Independence assumptions are also called *unconfoundedness* or *ignorability* assumptions in statistics, meaning ignorability of the assignment mechanism. Indeed, with independence, we don't need to know and model the assignment process to estimate causal effects, we need only compare group means. Earlier, we described the assignment mechanisms of random assignment and selection on observables solely for they imply what we are actually interested in for identification: the IA and the CIA, respectively.

¹¹ Δ The p -th QTE $Q_{Y^1}(p) - Q_{Y^0}(p)$ is the effect of the treatment at quantile p of the two marginal potential outcome distributions. It is not the p -th quantile of the individual treatment effect $Q_{Y^1 - Y^0}(p)$. In general, the latter quantile of a difference differs from the difference in quantiles. Independence identifies the marginal distributions of Y^0 and Y^1 , but not their joint distribution and hence not the distribution of $Y^1 - Y^0$. There is no similar problem in estimating the ATE, as $E[Y^1 - Y^0] = E[Y^1] - E[Y^0]$: differences in means equal means of differences.

In more general settings, what TE does the population OLS estimand recover?

We saw that when D is binary and there are no covariates X , (a) the population OLS slope equals the difference in observed conditional means: $\beta_{OLS} = \mathbb{E}[Y_i | D_i = 1] - \mathbb{E}[Y_i | D_i = 0]$, and (b) this in turn equals the ATET iff there is no selection bias. However, in more general settings, the population OLS estimand may diverge from the causal estimand of interest.

• With covariates X

Treatment effects are nothing more than differences in means. We could hence estimate them nonparametrically by comparing treated and untreated observations that share the same covariate values, as in exact matching. In practice, as the number of covariates increases, such nonparametric comparisons become increasingly difficult, so we use regression as a computational device. However, there are two main differences between the matching and the linear regression estimands:

1. An unsaturated (in X) linear regression imposes functional-form assumptions on the conditional expectations of the potential outcomes given X . In particular, it generally relies on a global linear approximation to the regression functions. Even if linearity is locally reasonable, a poor global approximation may produce substantial specification bias (Imbens and Wooldridge, 2009). It may therefore be preferable to combine regression with matching or weighting, so that regression is used only to adjust for remaining differences between relatively comparable observations.
2. Matching and OLS use different weights to aggregate the within-stratum differences in mean observed outcome $\delta_x := \mathbb{E}[Y_i | D_i=1, X_i=x] - \mathbb{E}[Y_i | D_i=0, X_i=x]$ —which, under the CIA, identify the CATEs $\mathbb{E}[Y_i^1 - Y_i^0 | X_i=x]$. If δ_x varies with x , i.e., if the TE is heterogeneous w.r.t. X , then the different sets of weights will result in different weighted averages of treatment effects (Angrist and Pischke, 2008, 3.3.1). For simplicity, consider a univariate and discrete X_i .

- In the exact-matching estimand, each stratum is weighted according to the conditional probability of treatment, which identifies the ATET.

$$\beta_M = \dots = \frac{\sum_x \delta_x w_M(x)}{\sum_x w_M(x)}, \quad w_M(x) := \Pr(D_i=1 | X_i=x) \Pr(X_i=x)$$

- In saturated OLS, each stratum is weighted according to the conditional variance of treatment, which is maximized when $\Pr(D_i=1 | X_i=x) = .5$, i.e., in strata with as many treated as control observations: OLS gives more weight to more balanced strata.

$$\beta_{OLS} = \dots = \frac{\sum_x \delta_x w_R(x)}{\sum_x w_R(x)}, \quad w_R(x) := \Pr(D_i=1 | X_i=x) (1 - \Pr(D_i=1 | X_i=x)) \Pr(X_i=x)$$

Hence, with heterogeneous treatment effects, the saturated OLS coefficient generally differs from the ATET. With an unsaturated linear specification for X , the OLS coefficient may additionally be affected by functional-form misspecification.

• With nonbinary D

We now consider a treatment with multiple levels of intensity. Let $D_i \in \{d^l, \dots, d^u\}$ denote treatment intensity, and $Y_i(d)$ denote individual i 's potential outcome under treatment level d —a dose-response function of d —and $Y_i := Y_i(D_i)$ their realized outcome. The generalized CIA and overlap conditions are $Y_i(d) \perp\!\!\!\perp D_i | X_i$ and $\Pr(D_i=d | X_i=x) > 0$ for all relevant values of (d, x) . For example, the within-stratum CATET for a 1-unit increase in treatment is then $\delta_x := \mathbb{E}[Y_i(d) - Y_i(d-1) | D_i=d, X_i=x]$.

- *Ideal case:* the dose-response function is linear in d and its slope does not vary with i : $\mathbb{E}[Y_i(d) | X_i] = \alpha + \beta d + X_i' \gamma$. Then a one-unit increase in treatment has the same causal effect β at every treatment level and for every x : $\mathbb{E}[Y_i(d) - Y_i(d-1) | X_i = x] = \beta$. Under the generalized CIA, the coefficient on D_i in the correctly specified regression $Y_i = \alpha + \beta D_i + X_i' \gamma + u_i$ identifies this common causal effect.
- *General case:* the dose-response function is nonlinear in d or its slope varies across i or X . Then the above regression estimates a weighted average of differences in the average dose-response function $Y_i(d) - Y_i(d-1)$, but it does not generally equal a particular ATET. One may instead include indicators for the treatment levels or estimate the dose-response function flexibly.

- **With limited Y**

We now consider an outcome that isn't continuous and unbounded, but is for example discrete or strictly positive.

In Angrist and Pischke (2008)'s view, the structure of the outcome variable is irrelevant, **linear regression** is always appropriate as the population linear projection **provides the best mean-square linear approximation to the conditional expectation function** (CEF) $\mathbb{E}[Y_i|D_i]$.¹⁵

- *Simplest case: binary D , no X .*

Because D_i takes only the values zero and one, its CEF is necessarily linear in D_i :

$$\mathbb{E}[Y_i | D_i] = \mathbb{E}[Y_i | D_i=0] + (\mathbb{E}[Y_i | D_i=1] - \mathbb{E}[Y_i | D_i=0]) D_i.$$

Thus, regardless of the constraints on the support of Y_i , we have $\beta_{OLS} = \mathbb{E}[Y_i | D_i = 1] - \mathbb{E}[Y_i | D_i = 0]$, which, absent selection bias, identifies the ATET.

- *General case: nonbinary D or additional covariates.*

The CEF $\mathbb{E}[Y_i|D_i, X_i]$ is generally nonlinear for limited Y_i (the saturated-covariate specification is impractical, and Y isn't normal so (Y, D, X) isn't multivariate normal). Linear regression then provides not the CEF itself, but still the best linear approximation to the CEF. As before, the CEF is causal under the CIA (i.e., identifies the conditional causal response function). Linear regression thus identifies the best linear approximation to the causal effect.¹⁶

¹⁵The question this raises however, is, in cases of a *highly nonlinear* CEF, what would we really learn from a linear approximation to it? A CEF is actually linear under only two rare circumstances: if (Y_i, D_i) is multivariate normal, or if the model is saturated (i.e., it has a separate parameter for each possible combination of values of the regressors).

¹⁶However, as the regression vector misses some features of the CEF, it would most likely generate fitted values outside the support of Y . This is for example a well-known problem of the linear probability model. For bounded or discrete outcomes, nonlinear conditional-mean models may fit the CEF better. But if one is interested only in estimating marginal effects (the average changes in CEF), this might be not so consequential.

2.2 The causal graph framework

An alternative to the potential outcomes framework for addressing causality has gained traction in disciplines other than economics: the causal graph framework, and more precisely the work on directed acyclic graphs (DAGs), largely developed by J. Pearl (Pearl, 2009). We introduce it here after noting two key points:¹⁷

1. The two frameworks are not opposed, they both define causality using counterfactuals — a causal effect is a comparison between two states of the world: a realized state as the causal variable took one value, and a counterfactual state that would have happened had it taken another value. The two frameworks then encode these counterfactual causal states differently.¹⁸
2. Each framework has benefits, which the next section attempts to summarize.

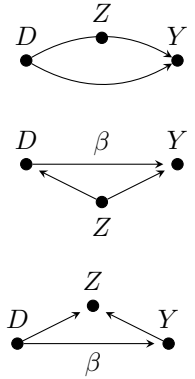
The potential outcomes and the causal graph frameworks are therefore complementary perspectives, and it can be useful to understand how to frame one’s causal analysis in the language of each.

2.2.1 Elements of directed acyclic graphs (DAGs)

Relationships are encoded with nodes and directed edges. Nodes represent random variables (circles are solid if they are observed, hollow otherwise), arrows represent possible direct causal relationships. A path is any sequence of edges. It is *closed* if at least one variable along the path is observed, *open* otherwise.

Three types of elementary paths can be sources of association between D and Y :

- **mediating path:** D causally affects Y through a mediator Z along a path.
 $\hookrightarrow$ Conditioning on Z would block this association, we would therefore recover only the direct causal effect of D on Y . Without conditioning, we would recover the total causal effect of D on Y .
- **confounding path:** a confounder Z determines both D and Y along a path.
 $\hookrightarrow$ Z creates a non-causal association between D and Y . Conditioning on it would block this association, we would therefore recover the ~~total association~~ causal effect β .
- **colliding path:** a collider Z is determined by both D and Y along a path.
 $\hookrightarrow$ Z creates no association between D and Y . Conditioning on it would induce a non-causal association between D and Y , we would therefore recover ~~the causal effect~~ β a non-causal association.

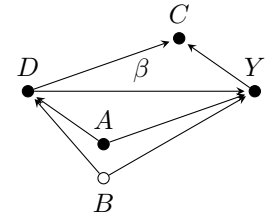


A **back-door path** is any path that begins with an arrow pointing to D and ends with one to Y .

Importantly, a DAG is a *complete* encoding of assumptions about causal relationships: those assumed to exist represented by arrows, and those assumed to not exist represented by missing arrows. I.e., the exclusion of an arrow is not a lack of assumption, but the assumption of no direct relationship: an exclusion restriction.

For example, the basic DAG on the right encodes:

- * explicitly, 4 paths linking D to Y :
 $D \xrightarrow{\beta} Y$: a direct (causal) path
 $D \leftarrow A \rightarrow Y$: a back-door confounding path, closed
 $D \leftarrow \cdot B \rightarrow \cdot Y$: a back-door confounding path, open
 $D \rightarrow C \leftarrow Y$: a colliding path
- * implicitly, 3 assumptions of no direct relationships between A , B and C .



2.2.2 Two main identification strategies: 1. Blocking back-door paths; 2. Instruments

We want to estimate the causal effect of a treatment D on Y . We represent in a DAG this causal relationship, and all other relationships relevant to the effect of D on Y . *Given the structure of the causal relationships,*

¹⁷For a detailed presentation, see Morgan and Winship (2015, ch. 1.5 & 3), of which this section is a proposed summary.

¹⁸How directed graphs encode causal states is not detailed here. See sections 3.4 and 3.6 of Morgan and Winship (2015), or Pearl (2009), for a detailed presentation. Importantly, we also consider only the subset of directed *acyclic* graphs (DAGs), where no directed paths emanating from a causal variable also terminate at the same causal variable. This prohibition of cycles notably rules out simultaneous causality and feedback loops. Section 3.2 of Morgan and Winship (2015) discusses the implications.

which variables must we observe and include to estimate the causal effect of D on Y ?

- Strategy 1: blocking back-door paths

The most common concern with observational data is that D and Y are partly determined by a third variable, i.e., that there is a back-door path. **The total association between D and Y equals β iff there are no back-door paths.**

- In the previous basic DAG:

- * Assume B wasn't there. The only back-door path between D and Y is closed as we observe A . If we adjust for A , i.e., hold it fixed, we remove the association between D and Y that is driven solely by fluctuations in A , and recover the causal effect β . **We can recover β by blocking all back-door paths**, i.e., conditioning on one confounder along each back-door path.

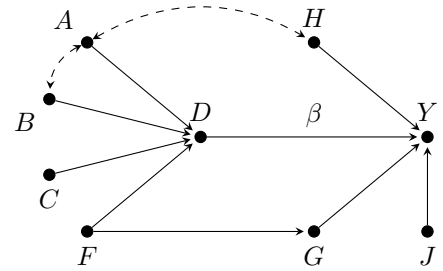
- * However, the back-door path through B is *open*, as B is unobserved. → We cannot recover β by blocking back-door paths.

- In the more complex DAG on the right, there are three back-door paths:¹⁹

$$\begin{aligned} D &\leftarrow A \leftrightarrow H \rightarrow Y \\ D &\leftarrow B \leftrightarrow A \leftrightarrow H \rightarrow Y \\ D &\leftarrow F \leftrightarrow G \rightarrow Y \end{aligned}$$

We can block all back-door paths by either:

- * conditioning on H and either F or G
- * conditioning on A and B ,²⁰ and either F or G ²¹



- Strategy 2: instruments

Instead of blocking back-door paths to estimate β directly, we can leverage an exogenous shock to D to estimate β indirectly. We use exogenous variation in an instrument Z ²² to isolate covariation in D and Y . In the DAG above, we can use as instrument for D either C , or F after conditioning on G .

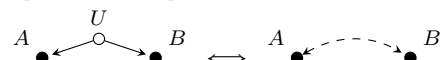
To estimate the effect β of D on Y , we reach the same conclusion as with the potential outcomes framework:

- In observational studies,
 - Leaving a back-door unblocked, i.e., excluding a confounding path, creates bias, so we must block all back-doors (adjust for all confounders).
 - *Back-door criterion*: With all back-doors blocked, i.e., all confounders conditioned on, we can recover an unbiased estimator of a causal effect.²³
 - Because we can rarely be certain that we have accounted for all confounders, we turn to alternative **identification strategies**, that rely on other assumptions.

2.3 Comparative strengths and weaknesses of the PO and DAG approaches

The potential outcomes (PO) framework being the most popular in econometrics, we ask what the DAG approach adds, and the ways in which it differs that are most relevant for work in econometrics.²⁴

¹⁹To show that two variables are mutually dependent on one or more unobserved common causes, instead of abiding by the definitions and showing it with U as in the left figure below, we can use a curved dashed bidirected edge as in the right figure as a shorthand. These bidirected edges should however not be interpreted as mere correlations between the two variables, they represent an unspecified set of unobserved common causes of the two variables that they connect.



²⁰Conditioning only on A would not suffice. As A is a collider along the path between B and H , conditioning only on A would create dependence between B and H , and so wouldn't eliminate the noncausal association between D and Y .

²¹We note here one insight from DAGs: we need not condition on F and G . The potential outcomes framework says one must adjust for all confounders, so we might think that we need to adjust for F and G . The DAG shows us that one suffices.

²²Instruments are formally introduced in the next section. In short, a variable Z is a valid instrument for D if it does not have an effect on Y except through its effect on D . We can then estimate consistently the effect of D on Y by taking the ratio of the relationships $Z \leftrightarrow Y$ and $Z \leftrightarrow D$.

²⁴Imbens (2020) proposes such a review, of which the majority of this section is a proposed summary.

• Role of experiments and manipulability

While the PO framework elevates randomized experiments as gold standard, the graphical literature doesn't deem experiments special. Related to that is the notion of manipulability:

- 👉 The PO framework defines the potential outcomes with reference to a manipulation, and thereby makes a distinction between attributes that are fixed for the units in the population, and causes that are manipulable. This implicit criterion of manipulability is potentially restrictive and unnecessary. The DAG literature does not deny causal character to nonmanipulable variables.
- 👉 Imbens (2020) finds it justified for economics as “policy relevance is a key goal.”²⁵ In any case, the conceptual framework of a manipulation has the benefit of clarifying the effect being identified.²⁶

• Parts of the causal analysis addressed

Consider the three parts of a causal analysis: 1. pre-identification: the development of a causal model; 2. identification: establishing whether a particular model is identified; 3. post-identification: statistical analyses (estimation and inference from a sample).

- Neither framework helps much with #1 (postulating a causal model of how the world works).
- The graphical literature considers the three steps as separate problems, and addresses almost only #2. We note that DAGs encode *nonparametric* causal relationships; no assumptions are made about the functional forms of dependence between variables and the variables' distributions. All interactions between the effects of different variables (e.g., D and X) on Y are also already permitted (directed edges to Y signify inclusion in the structural function $f_Y(D, X, \dots)$).
- On the contrary, in econometrics (based on the PO framework), most of the methodological literature on causality explicitly considers #2 and #3 jointly and is about estimation methods (e.g., the literature on weak instruments, propensity score...), as it sees many statistical problems in #3 as specific to the causal nature of the questions in #2.

• Representation of identifying assumptions and identification strategies

- 👉 The identifying assumptions that concern the existence or not of relationships are explicit in their graphical versions, and hence often much clearer than their algebraic versions. Ex: in IV settings, the DAG illustration of the exclusion restriction and independence assumption as missing arrows is arguably clearer than their expression as correlations between residuals and instrument.
- 👉 Other assumptions are not easily captured in the DAG framework, in particular shape restrictions (monotonicity, convexity...). Yet these play an important role in economic theory and applied identification strategies like IVs and RDs. The two main identifying assumptions in an RD setting are of a discontinuity in one conditional expectation and smoothness of other conditional expectations, and DAGs here arguably don't make them clearer (see Steiner et al. (2017)).
- 👉 The DAG literature proposes a machinery to infer identifiability given a complex model in a *systematic* way. In particular, it provides a criterion for choosing the variables to condition on — leading in identification strategies such as the backdoor criterion and the front-door criterion.²⁷ It shows the different ways to estimate a causal effect, and that “controlling for all other causes of Y ” can be misleading. For example, in the previous DAG #2, it showed that there were two possible strategies (after conditioning for either F or G): conditioning on either H or A and B .
- 👉 Accounting for treatment effect heterogeneity is difficult with DAGs, while the PO framework enables it. For example, the identification of LATEs is not easily derived in a graphical approach.

Aside from these theoretical concerns, Imbens (2020) suggests practical reasons for the lack of adoption of DAGs in economics:

²⁵He writes: “It is also not obvious to me why we would care [...] if the effect is not tied to an intervention we can envision.” and notes that much of the empirical work in economics focuses on questions about manipulations, e.g., “What would happen to my headache if I take an aspirin?” “How effective is a given treatment in preventing a disease?”

²⁶For example, consider a study comparing hiring outcomes when the racial-sounding of applicant names is changed. The conceptual framework of a manipulation helps clarify that the study cannot estimate the causal effect of race itself. What it captures is the well-defined causal effect of manipulation of the perception of race.

²⁷The front-door criterion is not developed here as it relies on exclusion restrictions that seem unrealistic in many social science applications. As Imbens (2020) points out, such difficulty in specifying credible models in economics “was a big part of the motivation for the so-called credibility revolution, with its focus on natural experiments.”

- The lack of concrete examples of the benefits of the DAG approach in realistic settings. The PO framework became popular because of empirical studies showing the merits of the proposed methods. *“In the absence of concrete examples that highlight their benefits over traditional methods, the toy models in the DAG literature sometimes appear to be a set of solutions in search of problems, rather than a set of clever solutions for substantive problems previously posed in social sciences.”*
- DAGs are by definition non-cyclical, and as such exclude questions of simultaneity and equilibrium behavior. Whereas equilibrium assumptions are central to economics, and are accommodated in the PO framework.

3 Design stage: canonical identification strategies

Hierarchy of common identification methods A contestable hierarchy of the most common identification methods in the ‘randomista’ toolkit, based on their capacity to mimic random assignment, is as follows:

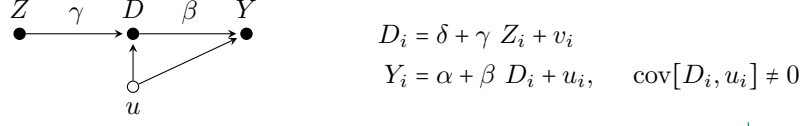
0. Randomized experiments (RCT) — or ‘natural’ randomization of treatment D
1. Instrumental variables (IV) and regression discontinuity (RD)
If there may be selection into treatment based on unobservables, we use an instrument or discontinuity that induces exogenous variation in treatment status.
2. Difference-in-differences (DiD), event studies and Synthetic control methods (SCM)
If we have repeated observations and want to estimate the effect of an event, we use research designs that assume or construct parallel trends and the presence of only time-invariant confounders.
3. Matching estimators
Strategies based solely on matching are considered much less credible — in terms of making us believe in the CIA, and thus their ability to recover a causal effect — than strategies based on some exogenous variation. Indeed, they require assuming that all confounders are observed (the assumed assignment mechanism is by selection on observables). While matching should therefore not replace a natural- or quasi-experiment design, it can however complement it. It is addressed in this context in section 4.

The sections below present, for each method and in its canonical setup: (i) the assumed data generating process (DGP), (ii) the identifying assumptions, (iii) the estimand, i.e., the treatment effect of interest, (iv) the estimator used, and (v) some best practices, and strengths and weaknesses. Importantly, the **relation between the actual observed outcomes Y_i and the conceptual potential outcomes Y_i^0, Y_i^1** is made explicit. This relation is crucial: it is the reason why our estimation (using Y_i) is able to recover a causal treatment effect (defined by Y_i^0, Y_i^1).

For simplification purposes, all methods are presented without the inclusion of exogenous controls X_i , but the relationships can be generalized to conditioning on covariates X_i .

3.1 IV

DGP $Y_i = \alpha + \beta_i D_i + u_i$, $\text{cov}[D_i, u_i] \neq 0$: D_i is endogenous. But $\exists$ a binary instrument Z_i that is a random source of variation in D_i , it *assigns* or changes the probability of treatment. We can use the instrument to isolate variation in D that is unrelated to u , and recover β .²⁸



Potential outcomes

We define the treatment assignment $Z_i \in \{0, 1\}$ and the treatment realization $D_i \in \{0, 1\}$. $Z_i = 0$ induces the potential treatment status D_i^0 , realized as 0 if individuals comply, 1 if not. $Z_i = 1$ induces D_i^1 , realized as 1 if they comply, 0 otherwise. The compliance behavior defines 4 categories of participants, which the researcher *cannot* observe; they can only observe Z_i and D_i .

	D_i^0	D_i^1
compliers	0	1
always-takers	1	1
never-takers	0	0
defiers	1	0

Identifying assumptions

- (A1) independence or ‘exogeneity’ (of Z w.r.t. the potential outcomes): $Z_i \perp\!\!\!\perp (Y_i^0, Y_i^1)$, which implies $\text{cov}[Z_i, v_i] = 0$. I.e., there is no unmeasured confounder that affects both the instrument and the outcome.
- (A2) exclusion restriction (no direct effect of Z on Y): Z affects Y only through D : $\text{cov}[Z_i, u_i] = 0$. $\rightarrow$ The effect of Z on Y or *intention to treat* of $ITT_{\text{never-takers}} = ITT_{\text{always-takers}} = 0$.
- (A3) relevance (of Z): $Z_i \not\perp\!\!\!\perp D_i$, which implies $\text{cov}[Z_i, D_i] \neq 0$
- (A4) monotonicity (of the effect of Z on D): Z is an incentive, it does not discourage treatment (no defiers). $\rightarrow$ Compliers are the only individuals whose treatment status could be altered by the instrument.

Estimand $\beta_{\text{IV}} := \frac{\text{cov}[Y_i, Z_i]}{\text{cov}[D_i, Z_i]} = \dots = \frac{\mathbb{E}[Y_i | Z_i=1] - \mathbb{E}[Y_i | Z_i=0]}{\mathbb{E}[D_i | Z_i=1] - \mathbb{E}[D_i | Z_i=0]} = \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 | D_i^0=0, D_i^1=1]}_{\text{LATE on the compliers}}$

Estimator A natural choice of estimator is the sample analog called *Wald estimator* $\hat{\beta}_W = \frac{\overline{\text{cov}}[Y_i, Z_i]}{\overline{\text{cov}}[D_i, Z_i]}$. $\hat{\beta}_W$ turns out to be numerically equivalent to the two-stage least squares (2SLS) estimator $\hat{\beta}_{2\text{SLS}}$ obtained through the two-step process:²⁹

1st stage: $D_i = \delta + \gamma \cdot Z_i + v_i \implies \widehat{D}_i = \widehat{\mathbb{E}}[D_i | Z_i] : \text{exogenous by construction}$

2nd stage: $Y_i = \tilde{\alpha} + \tilde{\beta} \cdot \widehat{D}_i + e_i$

Note: The reduced form of a model is that where the endogenous variables are expressed as functions of the exogenous variables. In the IV setting, the regression of Y on Z is therefore called the reduced form, and estimates the intention to treat (ITT), whereas the IV estimates the ATET.

Best practices

- Support the relevance assumption by showing a large F-statistic for the 1st stage (rule of thumb: $F > 10$). The bigger F , the *stronger* the instrument.
- As in any observational study, adjust for all other *relevant* pre-treatment variables (precisely, predictors of Y that would not be affected by D), making sure to include the same variables in both stages.

²⁸For more complicated treatment variables, we will need more complicated instruments. Ex: to identify *several* treatment variables, we will need at least as many instruments; to identify a *continuous* treatment, we can’t use a binary instrument.

²⁹The point estimates are equivalent, however the SEs of the 2nd stage would not give the correct SEs, as we need to adjust for the two stages of estimation. We must account for the estimation uncertainty from the first-stage (the first-stage is based on a sample, not the population, making $\widehat{D}_i$ a random variable, instead of the usual fixed variable). Most 2SLS packages do the adjustment automatically — otherwise one can simply bootstrap the SEs manually.

- Different valid instruments select different sets of compliers, leading to different estimands and thus estimates. Think of the group of compliers selected, to make sure the instrument is relevant w.r.t. the policy of interest. Then count and characterize these compliers to get more out of the LATE.
- For models that are non-linear in D , the properties of 2SLS do not necessarily hold, so one may want to consider alternative estimation strategies, such as the *control function method*:
 - Method: 2 stages: (i) same first stage, extract the residuals $\hat{v}$; (ii) OLS regression of Y on $(Z, D, \hat{v})$.
 - Limits: CF is generally more efficient but less robust than 2SLS as it imposes additional restrictions.

Strengths & weaknesses

- + Compelling identification strategy
- $\hat{\beta}_{IV}$ is less efficient than OLS, and this precision further decreases with weak instruments.
- $\hat{\beta}_{IV}$ has *finite sample bias*, which stems from the randomness in estimates of $\widehat{D}_i$ and increases with the weakness and number of instruments.
- ⇒ Beware of weak instruments. They can render $\hat{\beta}_{IV}$ considerably less efficient and even more biased than $\hat{\beta}_{OLS}$.³⁰ See Andrews et al. (2019).
- In many settings (e.g., models non-linear in D , non-saturated models with covariates), 2SLS can be very biased. New paper: <https://a-torgovitsky.github.io/tsls-weights.pdf>
- One can also use IVs to address the attenuation bias that may result from measurement error in D . Ex: Krueger and Lindahl (2001) uses an IV to address attenuation bias in cross-country estimates of the returns to education.

Counting and characterizing compliers to get more out of the LATE Compliers ($D_i^1 > D_i^0$) are rarely representative of the population, due to selective uptake. While we cannot identify individual compliers in the data, we can (i) estimate the size of the complier group, and (ii) characterize them in terms of their distribution of observed covariates.

- Counting compliers: We can measure (Angrist and Pischke, 2008, 4.4.4):
 - The size of the complier group. It's the Wald 1st stage: $\Pr(D_i^1 > D_i^0) = \dots = \mathbb{E}[D_i | Z_i=1] - \mathbb{E}[D_i | Z_i=0]$
 - The share of treated that are compliers:

$$\Pr(D_i^1 > D_i^0 | D_i=1) = \dots = \frac{\Pr(Z_i=1) \times (\mathbb{E}[D_i | Z_i=1] - \mathbb{E}[D_i | Z_i=0])}{\Pr(D_i=1)} = \frac{\text{share}(Z_i=1) \times \text{1st stage}}{\text{share treated}}$$

- Characterizing compliers: We can describe the distribution of covariates X for compliers.
 - For binary characteristics, we can calculate relative likelihoods (Angrist and Pischke, 2008, 4.4.4). For example, the likelihood that a complier has $X_i = 1$ relative to any individual is:

$$\frac{\Pr(X_i=1 | D_i^1 > D_i^0)}{\Pr(X_i=1)} = \dots = \frac{\mathbb{E}[D_i | Z_i=1, X_i=1] - \mathbb{E}[D_i | Z_i=0, X_i=1]}{\mathbb{E}[D_i | Z_i=1] - \mathbb{E}[D_i | Z_i=0]} = \frac{\text{1st stage} | X_i=1}{\text{1st stage}}$$

- For general covariates, we can calculate the mean —or other features of the distribution— of the covariate for compliers using Abadie (2003)'s kappa-weighting scheme: Suppose the identifying assumptions hold conditional on X_i . For any function $g(Y_i, D_i, X_i)$ with finite expectation, we have $\mathbb{E}[g(Y_i, D_i, X_i) | D_i^1 > D_i^0] = \frac{\mathbb{E}[\kappa_i g(Y_i, D_i, X_i)]}{\mathbb{E}[\kappa_i]}$, with the weighting function $\kappa_i = 1 - \frac{D_i(1-Z_i)}{1-P(Z_i=1|X_i)} - \frac{(1-D_i)Z_i}{P(Z_i=1|X_i)}$
- Applications: See Almond and Doyle (2011); Kowalski (2021).

To add:

³⁰Note that these are different biases: endogeneity bias with the OLS estimator and sample bias with the IV estimator.

- Shift-share instruments ([Borusyak et al., 2025](#)): They leverage two distinct sources of variation, by summing a common set of shifts with heterogeneous exposure share weights. Often used in studies of labor, trade, macroeconomics, public economics, finance. Ex: Shift = guerre en Ukraine, share = part des engrais qu'un autre pays importait d'Ukraine. Many economic studies consider units that are exposed differently to a common set of shocks.

Ex: Autor et al. (2013): How did the surge in Chinese imports in the 1990s and 2000s affect US local labor markets?

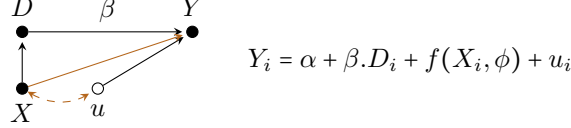
- X (= a shift-share explanatory variable) = “regional exposure to this China shock” = the extent to which workers were employed in industries that saw growing competition with China = the average of national industry-level shifts in US imports from China, weighted by the regional shares of employment across industries.
- Potential endogeneity concerns: US imports from China may be affected by US-specific productivity and demand shocks.
- IV (= a similar shift-share structure) = the average of industry growth in Chinese imports among non-US countries, again weighted by industry employment shares of US commuting zones.

3.2 RD

Known assignment mechanism but no overlap

Sharp RD

DGP Treatment D_i is not randomly assigned, it is deterministic, but *discontinuous* along a continuous pretreatment *running variable* X_i , s.t. there is *local randomization* around a cutoff c : $D_i = \mathbb{1}\{X_i \geq c\}$. Because D_i is a deterministic function of X_i , there are no confounding variables other than X_i . Given the trend relation $\mathbb{E}[Y_i^0 | X_i] = f(X_i)$, the DGP is described below, where the brown arrows disappear as $X \rightarrow c$:³¹



Δ There is zero overlap, as the assignment is a deterministic function of X_i (there is no value of X_i with both treatment and control observations), so we must extrapolate across X_i . We make comparisons by exploiting the continuity of potential outcomes across X_i . This means the RD estimate will be only as good as our model for $\mathbb{E}[Y_i^0 | X_i]$: we can't be that agnostic about functional form. But by looking only at data in a small neighborhood around c , the TE estimate should not depend much on this correct specification.

Identifying assumptions

- (A1) *local continuity*: the expected potential outcomes $\mathbb{E}[Y_i^1 | X_i]$ and $\mathbb{E}[Y_i^0 | X_i]$ are continuous in X_i at c . I.e., the other determinants of Y don't jump at c . $\implies$ The average outcome of those right below the cutoff (who are denied the treatment) are a valid counterfactual for those right above (who receive it).
- (A2) *relevance*: discontinuity in the dependence of D_i on X_i : $D_i = \mathbb{1}\{X_i \geq c\}$

Together, these assumptions imply that we can attribute a jump in Y_i at c to the causal effect of D_i .

Estimand $\beta_{RD} = \lim_{x \rightarrow c^+} \mathbb{E}[Y_i | X_i = x] - \lim_{x \rightarrow c^-} \mathbb{E}[Y_i | X_i = x] = \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 | X_i = c]}_{\text{LATE at the cutoff}}$

Estimator We can estimate β at the cutoff by running the regression:³²

$$Y_i = \alpha + \beta D_i + f(X_i) + e_i$$

Best practices

- Choice of $f()$: $f()$ is unknown. This is a problem, as misspecification of the functional form of the DGP may bias the estimator. The statistical challenge is of estimating a regression function nonparametrically at a boundary point. It is therefore done with flexible functional forms, such as:
 - a local linear regression model: $Y_i = \alpha + \beta D_i + \gamma_1(X - c) + \gamma_2(X - c)D + e_i$ with $c - h \leq X \leq c + h$.³³
 - a polynomial regression model with a low-degree polynomial, e.g., quadratic. Higher-order polynomials can lead to overfitting and introduce bias (Gelman and Imbens, 2019).

In both cases, report the results of several specifications to assess the sensitivity to $f()$.

- As in any observational study, adjust for all relevant pre-treatment variables. Just because the assignment depends on X , there is no reason to expect overlap and balance across pre-treatment characteristics not captured by X . We need to adjust for pre-treatment differences between the two groups.

³¹The causal graph is taken from [Steiner et al. \(2017\)](#).

³²To allow for different trend functions for $\mathbb{E}[Y_i^0 | X_i]$ and $\mathbb{E}[Y_i^1 | X_i]$ (i.e., to let the regression model differ on each side of the cutoff), add interactions between D and $f()$: $Y_i = \alpha + \beta.D_i + f(X_i, \phi_l) + f(X_i, \phi_r)D_i + e_i$

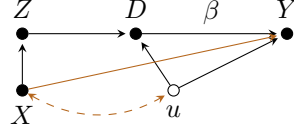
³³A larger bandwidth h increases precision but also bias. Choose the optimal h by estimating the model's predictive accuracy for different values of h , for example using leave-one-out cross-validation: iteratively for each observation i , fit the model using only the observations $X_i - h \leq \mathbf{X} < X_i < c$ when $X_i < c$, and only the observations $c < X_i < \mathbf{X} \leq X_i + h$ when $X_i \geq c$.

Strengths & weaknesses

- + RDDs are similar to a local randomized experiment, and thereby require weak assumptions.
- + RDDs are all about finding ‘jumps’ in the probability of treatment as we move along some X . They have much potential in economic applications, as geographic boundaries or administrative or organizational rules (e.g., program eligibility thresholds) often create usable discontinuities.
- They risk being underpowered.
- The parameter estimates are very *local*, their external validity may be low.

Fuzzy RD (imperfect compliance)

DGP At $X_i \geq c$ there is a jump, not in treatment assignment (D_i going from 0 to 1), but in the *probability* of treatment assignment $\Pr(D_i=1 | X_i)$. The discontinuity $Z_i := \mathbb{1}\{X_i \geq c\}$ becomes an instrumental variable for treatment status D_i . The DGP is represented in the causal graph below, where the brown arrows disappear as $X \rightarrow c$:



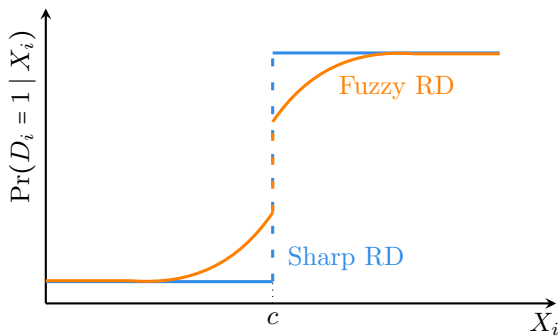
Estimand
$$\beta_{\text{RD}} := \frac{\lim_{x \rightarrow c^+} \mathbb{E}[Y_i | X_i=x] - \lim_{x \rightarrow c^-} \mathbb{E}[Y_i | X_i=x]}{\lim_{x \rightarrow c^+} \mathbb{E}[D_i | X_i=x] - \lim_{x \rightarrow c^-} \mathbb{E}[D_i | X_i=x]} = \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 | X_i = c]}_{\text{LATE at the cutoff}}$$

Estimator Fuzzy RD leads naturally to a simple 2SLS estimation strategy. The 2SLS estimator $\hat{\beta}_{\text{2SLS}}$ is obtained through the two-step procedure:

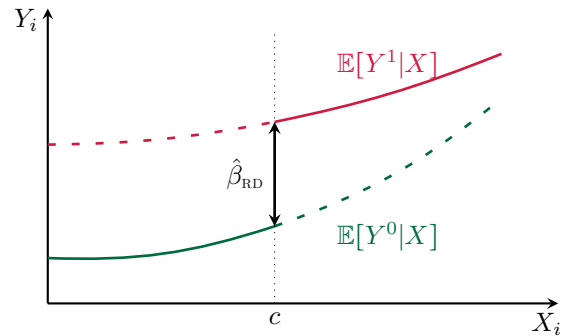
$$\text{1st stage: } D_i = \delta + \gamma \cdot Z_i + f(X_i) + u_i \implies \widehat{D}_i = \widehat{\mathbb{E}}[D_i | X_i]$$

$$\text{2nd stage: } Y_i = \alpha + \beta \cdot \widehat{D}_i + f(X_i) + e_i$$

As before, one can allow for treatment effects varying with X_i by adding treatment-covariate interactions.



(a) RD treatment assignment (sharp & fuzzy)



(b) RD treatment effect (sharp)

3.3 DiD, DiDiD, Event study, SCM

Repeated observations allow for adjusting for unobserved confounders in linear models

Repeated observations over some dimension allow for adjusting for all the unobserved characteristics that are *constant across that dimension*.

Say we have repeated observations for each individual i . Some characteristics stay constant for each individual (e.g., their birth place...). If before running our linear model, we transform each variable into its deviation from the group mean, i.e., we subtract the means for that individual, we remove any variation explained by these constant characteristics. We say we “control for individual” as we get rid of all the variation explained by the individual, i.e., the *variation between* individuals. We isolate the *variation within* each individual. The *within* estimator compares each individual to themselves (across the dimension over which we have the repeated observations).^a

There are two standard estimation methods to do this:

- Option #1: De-meaning manually: $Y_{it} - \bar{Y}_i = \alpha_0 + \beta(X_{it} - \bar{X}_i) + u_{it}$
– This gets complicated if there are multiple dimensions along which to remove variation.
- Option #2: Adding *individual fixed effects*, i.e., a set of binary indicators as explanatory variables, one for each individual: $Y_{it} = \alpha_i + \beta X_{it} + v_{it}$
– This is easy to run with most statistical softwares, but can be computationally intensive as it estimates an intercept for each individual (even though we won’t interpret them).

We can also include multiple sets of fixed effects to adjust for multiple sets of unobserved confounders. What are the comparisons that would be averaged into β ? For example, consider repeated cross-sectional data of groups $g = 1, \dots, G$ over periods $t = 1, \dots, T$:

- w. group FEs: β = average of TEs that compare group peers (among themselves and across t);
- w. period FEs: β = average of TEs that compare observations within the same period;
- w. both (*two-way fixed effects*): β = average of TEs identified from (i) variation within group and (ii) variation within period.^b

Note finally that as the OLS estimator is a weighted average of treatment effects, where weights are proportional to the conditional variance of treatment, it will weight a lot more the units with large *within* variation in treatment.

^aWe could instead decide to de-mean at a higher level, e.g., group. We would then be comparing individuals in each group only to other individuals in that group.

^bEach comparison is relative to what is expected given that group, and given that year. Note that this is not the same as ‘given that group that year’ (such comparisons would rely on isolating variation within group-year, which would be obtained with group-by-year fixed effects). Here, each of the ‘relative to’ stands alone.

We consider in this section the setting of a treatment assigned at a certain time, and units observed before and after the assignment, i.e., repeated observations over time. To estimate the causal effect of the event, we need a model to estimate the counterfactual value (the unit’s outcome if the event had not occurred). The sections below present **three different models of that counterfactual**:

- **DiD, DiDiD** Some units never get treated. Assuming that counterfactual trends (Y^0) of the treated and untreated are parallel, we can use the control units to remove trends in Y in the treated. We identify effects from both *within* and *between* variation.
- **Event studies** All units get treated. Assuming that a unit’s past value is a plausible counterfactual value, we identify effects from *within* variation only.
- **SCM** Units are large aggregates, and most never get treated. Assuming parallel counterfactual trends across these individual units is unreasonable. But we can combine them to construct an optimal synthetic unit that captures the counterfactual trajectory of the treated unit(s).

DiD

DGP Treatment assignment or exposure is a function of two dimensions: group (treatment/control) and most commonly time (pre/post exposure at τ).³⁴ We define the associated binary variables $G_i := \mathbb{1}\{i \in \text{treatment group}\}$ and $P_t := \mathbb{1}\{t \in \text{post period}\} = \mathbb{1}\{t \geq \tau\}$.

- In a before/after comparison within the treatment group, the difference in Y could result from other changes that occur during the time period (time effects)...
- In a treatment/control group comparison within the post period, the difference in Y could result from permanent differences between the groups (time-invariant confounders)...
- We can remove both biases by comparing the *change over time in $\bar{Y}$* in the treatment group to the *change over time in $\bar{Y}$* in the control group.

Identifying assumptions

- (A1) Parallel counterfactual trends across groups: in the absence of treatment, both groups would have experienced the same *changes (pre $\rightarrow$ post) in outcomes Y_{it}* : $\mathbb{E}[Y_{i1}^0 - Y_{i0} \mid G_i=1] = \mathbb{E}[Y_{i1}^0 - Y_{i0} \mid G_i=0]$.
This excludes selection into treatment based on differential outcome trends.
- (A2) Stable group compositions over time.

Estimand $\beta_{\text{DiD}} := \left(\mathbb{E}[Y_{i1} \mid G_i=1] - \mathbb{E}[Y_{i0} \mid G_i=1] \right) - \left(\mathbb{E}[Y_{i1} \mid G_i=0] - \mathbb{E}[Y_{i0} \mid G_i=0] \right) = \dots = \underbrace{\mathbb{E}[Y_{i1}^1 - Y_{i1}^0 \mid G_i=1]}_{\text{ATET in post period}}$

Estimator The OLS estimator $\hat{\beta}_{\text{OLS}}$ of the following saturated regression consistently estimates β_{DiD} :

$$Y_{it} = \alpha + \beta_G G_i + \beta_P P_t + \beta G_i P_t + e_{it}$$

In this 2-groups 2-periods design, β_{DiD} is equal to the treatment coefficient in a two-way fixed effect (TWFE) regression with group and period fixed effects: $Y_{it} = \lambda_G + \lambda_P + \beta G_i P_t + u_{it}$ or $Y_{it} = \lambda_G + \lambda_t + \beta G_i P_t + v_{it}$.

Best practices

- Support the assumption of parallel counterfactual trends by testing its pre-trends implication (if we have data for multiple pre-periods). Estimate the following *event-study* aka *dynamic TWFE* regression model by OLS:

$$y_{it} = \lambda_G + \lambda_t + \sum_{t \neq \tau-1} \beta_t G_i \mathbb{1}\{t\} + e_{it}$$

$\forall t, \beta_t$ is the *treated-control* outcome gap in period t relative to the *treated-control* gap in the reference period $\tau-1$, i.e., $\forall t < \tau-1$, it measures whether treated and control units were already drifting apart before treatment, relative to the last pre-period.

$\rightarrow$ Report the joint Wald/F-test of $H_0 : \{\beta_t\}_{t < \tau-1} = 0$.³⁵

- The regression above also enables us to look at the cumulative TEs over time: $\beta_{t, t \geq \tau}$ is the cumulative treatment effect up to period t , relative to $\tau-1$ (and, e.g., β_{tau} is the immediate one-period DiD estimate).
- If the composition of the groups changes over time, interact covariates X with P_t .
- As in any observational study, adjust for all other relevant pre-treatment variables.

³⁴In the canonical DiD setting, the second dimension is time, but it need not be. Data could be grouped by cohort (e.g., year of birth) or other characteristics.

³⁵Testing separately whether each of the $\{\beta_t\}_{t < \tau-1}$ is individually significant is not a valid test of the parallel trends assumption as it ignores multiple testing, individually insignificant coefficients can still be jointly significant, and parallel trends is a global restriction across all pre periods. It is useful to show the event-study plot with CIs for transparency, but the relevant hypothesis test is of the joint null that all pre-treatment event-time coefficients equal zero.

Strengths & weaknesses

- + Repeated observations get rid of unobserved time-invariant confounders, creating comparable groups.
- + Pre-trends aren't a problem (unlike in event-studies) as long as those of the two groups are *parallel*.
- + Identification only requires repeated observations, so repeated cross-sectional data suffice, as long as the sample composition does not vary over time. Panel data satisfy this condition by construction.

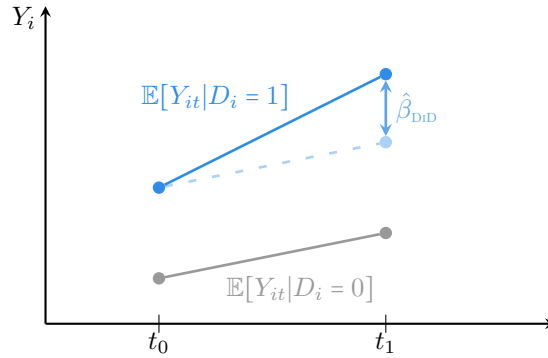


Figure 2: DiD treatment effect

DiDiD

DGP The treatment varies along a 3rd dimension or *subgroup*, e.g., gender, space... We define the binary variable $S_i := \mathbb{1}\{i \in \text{treatment in dimension \#3}\}$.

Identifying assumptions

(A1) Same counterfactual trends across ~~groups~~ subgroups: in the absence of treatment, the difference in subgroups would have experienced the same *changes (pre → post) in outcomes*:

$$\mathbb{E}[Y_{i1}^0 - Y_{i0} | G_1, S_1] - \mathbb{E}[Y_{i1}^0 - Y_{i0} | G_1, S_0] = \mathbb{E}[Y_{i1}^0 - Y_{i0} | G_0, S_1] - \mathbb{E}[Y_{i1}^0 - Y_{i0} | G_0, S_0]$$

(A2) Stable subgroup compositions over time.

Estimand

$$\begin{aligned} \beta_{\text{DiDiD}} &:= \left[(\bar{Y}_{G_1 S_1 P_1} - \bar{Y}_{G_1 S_1 P_0}) - (\bar{Y}_{G_0 S_1 P_1} - \bar{Y}_{G_0 S_1 P_0}) \right] - \left[(\bar{Y}_{G_1 S_0 P_1} - \bar{Y}_{G_1 S_0 P_0}) - (\bar{Y}_{G_0 S_0 P_1} - \bar{Y}_{G_0 S_0 P_0}) \right] \\ &= \dots = \underbrace{\mathbb{E}[Y_{i1}^1 - Y_{i1}^0 | G_i=1, S_i=1]}_{\text{ATET in post period}} \end{aligned}$$

Estimator The OLS estimator $\hat{\beta}_{\text{OLS}}$ of the following regression consistently estimates β_{DiDiD} :

$$Y_{it} = \alpha + \beta_G G_i + \beta_S S_i + \beta_P P_t + \beta_{GS} G_i S_i + \beta_{GP} G_i P_t + \beta_{PS} P_t S_i + \beta_{GSP} G_i S_i P_t + e_{it}$$

Best practices

- A triple difference makes for a very specific control group. Before doing an DiDiD, one should explain why a double difference isn't satisfactory (i.e., why the control and treatment groups in the double difference may have different counterfactual trends), and even the first difference.
- As in any observational study, adjust for all other relevant pre-treatment variables.

Strengths & weaknesses

- + A triple difference can difference out more confounding elements, hence it is harder to find confounders.
- It requires more data and variation.

ADD: example and figure for DiDiD: see Joan Llull's lecture notes: http://pareto.uab.cat/jllull/BGSE_Quantit_Meth/Class_Notes_QSMII_Chapter_7.pdf

Event study

DGP We want to estimate the causal effect of *an event*, which occurs for *all units* in the population but at different times τ_i , on some outcome Y . Treatment assignment is a function of the period (pre/post τ_i).

Identifying assumptions

- (A1) Conditional exogeneity of the timing of the event: Conditional on controls, the timing of the event is independent of potential outcomes.³⁶ Parallel counterfactual trends (i.e., absent the event, outcomes would have evolved similarly across event times) is an implication of this assumption. We can then reasonably use the group's past outcome value to construct its counterfactual post-event outcome value.
- (A2) Stable sample composition: the set of observed units does not change differentially around the event in a way correlated with treatment status (i.e., no endogenous selection).

Estimand $\beta_{ES} := \mathbb{E}[Y_{it} | t=\tau_i] - \mathbb{E}[Y_{it} | t=\tau_i-1] = \mathbb{E}[Y_{i,\tau_i}^1] - \mathbb{E}[Y_{i,\tau_i-1}] = \mathbb{E}[Y_{i,\tau_i}^1] - \mathbb{E}[Y_{i,\tau_i}^0] = \underbrace{\mathbb{E}[Y_{i,\tau_i}^1 - Y_{i,\tau_i}^0]}_{\text{ATET}}$

Estimator The OLS estimator $\hat{\beta}_{OLS}$ of the following regression (including unit fixed effects, calendar-time fixed effects, and event-time indicators—omitting the period before the event to normalize it as 0, s.t. the normalized *event time* variable captures the number of periods since treatment) consistently estimates β_{ES} :

$$Y_{it} = \sum_{t'=-K}^{\tau_i-2} [\beta'_t \mathbb{1}\{t=t'\}] + \beta \mathbb{1}\{t=\tau_i\} + \sum_{t'=\tau_i+1}^L [\beta'_t \mathbb{1}\{t=t'\}] + \alpha_i + \delta_t + e_{it}$$

This specification can be called *dynamic*, in that the coefficients $\{\beta_t\}$ capture dynamic treatment effects: the effects are allowed to change non-parametrically over time since the event.

Best practices

- Test in support of (A1): Plot/report all β_t s, to check that they are not changing up to the event. A change would suggest the presence of pre-trends, which makes it hard to interpret the event (unless there is a sharp trend discontinuity) as they suggest some endogeneity of D .
- Test in support of (A2): E.g., in the case of an unbalanced panel where outcomes aren't measured periodically, test for monitoring endogeneity by regressing the probability of observing the outcome on treatment and controls.
- As in any observational study, adjust for all relevant pre-treatment variables.

Strengths & weaknesses

- It is difficult to rule out other things changing at the same time, i.e., unobserved confounders (time-varying unobservables correlated with both D_{it} and Y_{it}).

⚠ Two-Way Fixed Effects estimators can be unreliable with heterogeneous TEs

Assuming identification assumptions hold, the estimation of a weighted ATET is very simple in the ideal setting of a binary D that only switches on, and is assigned at the same time for all units. However, in more complicated settings (where D is nonbinary, staggered, switches off...), naive extensions of these methods estimate quantities that can be very far from an informative ATET.

Let's assume that the parallel trends assumption holds. We saw that in the canonical 2-groups 2-periods setting, $\hat{\beta}_{DID}$ is equal to the two-way fixed effects (TWFE) estimator $\hat{\beta}_{FE}$; both estimators are unbiased for the ATET. However, in designs with more variety in exposure to treatment (with many groups and periods, staggered treatment, treatment switching off, non-binary treatments...), the TWFE estimator may be biased

³⁶Formally, $\mathbb{E}[e_{it} | \{D_{is}\}_{s \geq t}, \alpha_i, \delta_t, X_{it}] = 0$. This expression for $s = t$ corresponds to *contemporaneous* conditional exogeneity, which rules out correlation between current shocks and current unobservables. The case where $s > t$ covers *dynamic* conditional exogeneity, which rules out correlation of current unobservables with future realizations of the treatment, i.e., corresponds to the assumption of no anticipation.

if TEs are not constant across groups or over time (e.g., a policy becoming more or less effective). I.e., even with all confounders accounted for, $\hat{\beta}_{FE}$ is not robust to heterogeneity of TEs across groups or periods.

1. Source of the problem

Consider the TWFE regression model: $Y_{it} = \alpha_{g[i]} + \gamma_t + \beta_{FE} D_{g[i]t} + e_{g[i]t}$, for unit i in group g at time t . $\hat{\beta}_{FE}$ is a specific weighted sum of the ATEs in each treated (g, t) cell, with each weight $\mathbf{w}_{gt}$ proportional to and of the same sign as $N_1(D_{gt} - D_{g..} - D_{.t} + D_{..})$ and $\sum_{g,t} \mathbf{w}_{gt} = 1$,³⁷ such that in general, $\mathbb{E}[\hat{\beta}_{FE}] \neq \beta_{ATET}$ (de Chaisemartin and D’Haultfoeulle, 2020):

$$\beta_{ATET} = \mathbb{E} \left[\sum_{(gt): D_{gt}=1} \frac{N_{gt}}{N_1} \text{ATE}_{gt} \right], \quad \mathbb{E}[\hat{\beta}_{FE}] = \mathbb{E} \left[\sum_{(gt): D_{gt}=1} \frac{N_{gt}}{N_1} \mathbf{w}_{gt} \text{ATE}_{gt} \right]$$

- ✓ In the textbook case (D is binary, only switches on, and is assigned at the same time for everyone), $D_{gt} - D_{g..} - D_{.t} + D_{..}$ is constant across (g, t) cells, therefore $\hat{\beta}_{FE}$ is unbiased for the ATET.
- ✗ In more complicated settings, the $\mathbf{w}_{gt}$ vary. Then, heterogeneity in ATE_{gt} leads to a biased $\hat{\beta}_{FE}$. Some $\mathbf{w}_{gt}$ may even be negative (i.e., $\hat{\beta}_{FE}$ may not even identify a convex combination of TEs).³⁸ One-way FE regressions may similarly be biased for the ATE, but unlike TWFE regressions they always estimate a convex combination of TEs.
 - When D is binary, staggered, in static TWFE regressions, $\hat{\beta}_{FE}$ is a weighted average of all possible 2-group, 2-period DiD estimators in the data, where each weight is a function of the sample size and the subsample variance of treatment (Goodman-Bacon, 2021).³⁹ Some of these 2x2 DiDs misuse an early-treated group as control for a late-treated group, which may induce negative weights if the TE varies over time.
 - When D isn’t binary, β_{FE} may leverage another type of comparison: it may compare the outcome evolution of a group whose treatment increases more to the outcome evolution of a group whose treatment increases less.
 - When D is binary, staggered, in dynamic TWFE regressions, the coefficient on a given lead or lag can be contaminated by effects from other periods, and thus is not a valid estimate of a dynamic treatment effect TE heterogeneity, and apparent pre-trends can misleadingly arise solely from TE heterogeneity (Sun and Abraham, 2021).

Wooldridge (2021) highlights that the cause of the problem is not the TWFE estimator per se, but its misuse: it is applied to a restrictive model, namely, which does not allow for heterogeneity in the TE.

2. Alternatives

de Chaisemartin and D’Haultfoeulle (2022) summarizes the fast-growing literature on this problem, and highlights:

- Diagnosis tools
 - ▶ The `twowayfeweights` command (in R and Stata) computes the weights $\frac{N_{gt}}{N_1} \mathbf{w}_{gt}$.
 - ▶ The `bacondecomp` command (in R and Stata) computes the 2x2 DiD estimators and weights entering in β_{FE} , in the case of a binary staggered D .
- Alternative estimators

Ex: Wooldridge (2021) proposes an *extended TWFE* approach (based on the random-effects Mundlak estimator) which notably interacts the treatment indicator with time and/or group-time dummies to allow TEs to change across groups or periods.

³⁷Where N_{gt} is the number of observations in cell (g, t) ; N_1 is the total number of treated observations; a dot subscript means the variable’s average is taken over the given dimension; and while the original demonstration considers a binary treatment, the results “apply to any ordered treatment” (de Chaisemartin and D’Haultfoeulle, 2022), s.t. $\text{ATE}_{gt} = \frac{Y_{gt}^{D_{gt}} - Y_{gt}^0}{D_{gt}}$. Precisely, $w_{gt} = \frac{\tilde{e}_{gt}}{\sum_{(gt): D_{gt}=1} \tilde{e}_{gt} N_{gt} / N_1}$, where $\tilde{e}_{gt}$ is the residual in the regression of D_{gt} on group and period FEs.

³⁸This can even lead to a negative coefficient $\hat{\beta}_{FE}$ while the true ATEs are positive for everyone. Ex: $1.5 \times 1 - 0.5 \times 4 = -0.5$.

³⁹OLS will give more weight to subgroups where the FE-adjusted treatment dummy varies more. As a result, the timing of a unit’s treatment will determine its weight in the regression. If i is treated very early or very late, then it will have very little variation in treatment across the period (it is 0 almost the whole time, or 1 almost the whole time) and so will receive little weight. Note that weighting by treatment-variance is how OLS handles heterogeneity all the time — see section 2.1.2.

More generally, since the source of the ‘problem’ is the heterogeneity in TE, we should be thinking about how to allow for heterogeneity in our model.

SCM

DGP An event or intervention occurs at the level of aggregate entities (e.g., cities, countries) at time $t = \tau$. A single unit $j = 1$ is affected, the other units $j = 2, \dots, J + 1$ are unaffected. As the units of observation are a small number of aggregate entities, no single unaffected unit may provide a good comparison for the exposed unit. But a combination of unaffected units could. For each unit j , we observe a set of k predictors of the outcome, $X_{1j}, \dots, X_{kj}$ which include pre-intervention values of Y_{jt} and are themselves unaffected by the intervention.

We consider a *synthetic control unit*, which is a weighted average of the different J units in the donor pool that should reproduce the behavior of Y for the treated unit in the absence of the treatment. We call *synthetic control* the set of these optimal weights $W^* := (w_2^*, \dots, w_{J+1}^*)$.

Identifying assumption

- (A) The synthetic control unit reproduces the behavior of Y for the treated unit 1 in the absence of the treatment.

Estimand $\beta_{\text{SCM},t} := Y_{1t} - \hat{Y}_{\text{su},t} = \underbrace{Y_{1t}^1 - Y_{1t}^0}_{\text{TET}}$

Estimator The synthetic control estimator:

$$\forall t, \hat{\beta}_{\text{SCM},t} := Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt}$$

⚠ The synthetic control estimator is not regression-based, so we don't also get a standard error / p-value. After estimating the treatment effect, we need to do additional steps to generate inferences from that estimate.

Best practices / Steps

1. Design phase: Creating the comparison unit = determining the optimal weights W^*

- Common approach: The weights should minimize the distance between the treated unit and the synthetic unit, in terms of ability of predicting the post-treatment outcomes. We therefore calculate the weights that minimize the distance between the vector of pre-treatment characteristics for the treated unit X_1 and the matrix of pre-treatment characteristics for the untreated units X_0 :

$$\begin{aligned} w_j^*(V) &:= \underset{w_j}{\operatorname{argmin}} \quad \|X_1 - X_0 W\| = \sqrt{(X_1 - X_0 W)' V (X_1 - X_0 W)} \\ \text{s.t.} \quad &w_j \geq 0 \quad \forall j = 2, \dots, J + 1 \\ &\sum w_j = 1 \end{aligned}$$

where $V_{k \times k}$ represents the relative importance (weight) to give to each variable as a predictor of Y^0 . For example, if $V_{k \times k}$ is diagonal with elements $v_1, \dots, v_k$, the minimization function simplifies to $w_j^* := \underset{w_j}{\operatorname{argmin}} \sum_{m=1}^k v_m \left(X_{1m} - \sum_{j=2}^{J+1} w_j X_{jm} \right)^2$.

To solve this, first, we need to choose the set of predictors $X_1, \dots, X_k$ and the matrix V .

- *Choosing $X_1, \dots, X_k$ and V* should be based on their predictive value. One often-used selection criterion is the mean squared prediction error (MSPE). One divides the pre-treatment period into an initial training period and a subsequent validation period, computes SC weights using the training data, and evaluates the predictive power of the resulting synthetic control using the validation data, by computing the MSPE. One then chooses the synthetic control $W(V)$ that minimizes this MSPE w.r.t. $Y_{1,t < \tau}$, for some set of pre-treatment periods: $\text{MSPE} = \sum_{t < \tau} \left(Y_{1t} - w_2 Y_{2t} - \dots - w_{J+1} Y_{J+1,t} \right)^2$.

- Other approaches include best subset regression or LASSO and elastic nets methods, which perform better in settings with a large number of potential control units (see Doudchenko and Imbens (2016)).

2. Estimation phase

- Report the weights $w_1^*, \dots, w_{J+1}^*$.
- Check that the synthetic unit has values similar to the treated unit across all matching variables.
- Show robustness to:
 - (i) the choice of units $j = 2, \dots, J + 1$ in the donor pool. Ex: leave-one-out re-analysis: take from the sample one-at-a-time each of the units that contribute to the synthetic control, and check that the estimate is robust to the exclusion of any particular unit.
 - (ii) the choice of predictors $X_1, \dots, X_k$.

3. Inference phase: Is the treatment effect on the exposed unit *significant*? I.e., is it in the tails (extreme) of some distribution of treatment effects?

- Compute the exact p-value through placebo-based inference
 - (a) Generate a distribution of placebo effects by iteratively reassigning the treatment to each untreated unit, and applying the SCM.
 - (b) Choose a test statistic, and calculate it for each placebo. Abadie et al. (2010) recommends calculating the ratio of the post-treatment root mean squared prediction error (RMSPE) to the pre-treatment RMSPE.
 - (c) Sort the values of this statistic in descending order.
 - (d) Calculate the p-value (the treatment unit's statistic in the distribution) as $p := \frac{\text{rank}}{\text{total} = J}$.

One can plot a histogram of the statistics, and mark the treatment unit/group in the distribution to show the exact p-value associated with the model. One can also plot all the placebos, and highlight the treated unit's trajectory.

- Construct confidence intervals through test inversion (see Firpo and Possebom (2018))

Strengths & weaknesses

- + The SCM formalizes the creation of a comparison unit using a data-driven procedure. The *synthetic unit* is an optimally estimated counterfactual to the unit that received the treatment. The approach combines elements from matching and DiD techniques.
- + Transparency
 - of the fit: By showing the actual discrepancy between the treated unit and the convex combination of untreated units: for each of the X chosen (= balance table), and in terms of pre-intervention outcomes (through the graph of the two times series).
 - of the counterfactual: The weights chosen make explicit what each unit is contributing to the counterfactual. Whereas regression also weights the data, but one cannot see the weights.
- + It is a generalization of DiD, which relaxes the parallel trends assumption, by instead building an artificial control to have the best possible pre-trend.
- + It precludes extrapolation beyond the support of the data (which can occur in extreme situations with regression), by restricting the weights to be non-negative and to sum to 1.
- Requires additional steps to generate inferences.
- ⚠ SC doesn't solve the problem of subjective researcher bias. Through the choice of the covariates themselves, one can in principle select different weights.⁴⁰

⁴⁰Ferman et al. (2020) shows there remains possibilities for specification searching in the SC method... The authors consider a variety of commonly used SC specifications (e.g., all pre-treatment y values, the first 3/4 of the pre-treatment y values...), run the randomization inference test to calculate empirical p-values, and find that the probability of falsely rejecting the null in at least one specification for a 5% significance test can be as high as 14%. I.e., it is theoretically possible to 'hack' the analysis in order to find statistical significance that suits one's priors, by specification searching. This happens even when the number of pre-periods is very high. Restricting the number of pre-periods (as an attempt to limit the ability to conduct an endogenous specification search) isn't a good solution either as it diminishes control for unobserved confounders (it misallocates more weights). $\implies$ SC doesn't remove subjective researcher bias, despite creating weights based on a data-driven optimization algorithm. The

Extensions

- Synthetic Difference-in-Differences ([Arkhangelsky et al., 2021](#))
- More recently, the SCM has been applied to settings with a large number of units.
- Relaxing functional form assumptions. Ex: it isn't clear whether one should assume that it is the *percentage change* or the *absolute change* in the level of average outcomes over time that would have been the same across groups in the absence of treatment. For the case of repeated cross-sectional data, [Athey and Imbens \(2006\)](#) proposes a nonlinear version of the difference-in-differences model, called *changes-in-changes*, which does not rely on functional form assumptions, while still allowing the effects of time and treatment to vary systematically across individuals. It provides the distribution of outcomes, beyond the average effect of the treatment itself.

weights are optimal in that they uniquely minimize the distance function, but the distance function is endogenously chosen by the researcher. It is recommended to present multiple results under a variety of commonly specified specifications, instead of one specification which may be the cherry-picked result.

Summary of canonical identification strategies, and regression-based consistent estimators

Method	Identifying assumptions	Estimand β nd corresponding TE	Chosen estimator $\hat{\beta}$	Strengths / Weaknesses
RCT	(A) independence	$\beta := \mathbb{E}[Y_i D_i=1] - \mathbb{E}[Y_i D_i=0] = \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0]}_{\text{ATE}}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_{it} = \alpha + \beta D_i + e_{it}$. Consistent and unbiased.	👉 Random assignment guarantees (A) ⇒ RCT = “gold standard”
Selection on observables	(A1) conditional independence (A2) overlap	$\begin{aligned} \beta(x) &:= \mathbb{E}[Y_i D_i=1, X_i=x] - \mathbb{E}[Y_i D_i=0, X_i=x] \\ &= \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 D_i=1, X_i=x]}_{\text{CATE, } x} \\ \beta &:= \sum_x \frac{w(x)}{\sum_x w(x)} \beta(x), \quad w(x) := \mathbb{P}[D_i X_i=x]P(X_i=x) \end{aligned}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_{it} = \alpha + \beta D_i + f(X_i) + e_{it}$. Consistent, unbiased if $f(\cdot)$ is correct.	👉 hard to know all confounders are observed 👉 need globally accurate assumption of the forms $f(\cdot)$ of $Y_i^1 X_i, Y_i^0 X_i$
IV	Exogenous variation in D induced by Z . (A1) independence (A2) exclusion restriction (A3) relevance (A4) monotonicity	$\begin{aligned} \beta_{\text{IV}} &:= \frac{\text{cov}[Y_i, Z_i]}{\text{cov}[D_i, Z_i]} = \dots \\ &= \frac{\mathbb{E}[Y_i Z_i=1] - \mathbb{E}[Y_i Z_i=0]}{\mathbb{E}[D_i Z_i=1] - \mathbb{E}[D_i Z_i=0]} : \text{“Wald estimand”} \\ &= \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 D_i^1=1, D_i^0=0]}_{\text{LATE, compliers}} \end{aligned}$	$\hat{\beta}_{\text{W}} := \frac{\widehat{\text{cov}}[Y_i, Z_i]}{\widehat{\text{cov}}[D_i, Z_i]} = \dots =$ numerically equivalent to $\hat{\beta}_{\text{2SLS}}$ of the 2-stage regression $D_i = \delta + \gamma \cdot Z_i + v_i$ $Y_i = \tilde{\alpha} + \tilde{\beta} \cdot \hat{D}_i + e_i$ Consistent, biased , bias ↓ w. Z strength	👉 reliable exogenous variation 👉 less efficient than $\hat{\beta}_{\text{OLS}}$ if instrument is weak
Sharp RD	Discontinuity in treatment assignment via a cutoff in X . (A1) local continuity (A2) relevance	$\begin{aligned} \beta_{\text{RD}} &:= \lim_{x \rightarrow c^+} \mathbb{E}[Y_i X_i=x] - \lim_{x \rightarrow c^-} \mathbb{E}[Y_i X_i=x] \\ &= \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 X_i=c]}_{\text{LATE at the cutoff}} \end{aligned}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_i = \alpha_l + \beta D_i + f(X_i) + e_i$, with choice of $f(\cdot)$: 👉 local linear regression 👉 polynomial regression Consistent, biased , bias ↑ w. bandwidth	👉 akin to a local randomized experiment 👉 rather testable assumptions 👉 risks being underpowered 👉 maybe low external validity
Fuzzy RD	Discontinuity in $\Pr(\text{treatment assignment})$ via a cutoff in X . (A1) local continuity (A2) relevance	$\begin{aligned} \beta_{\text{RD}} &:= \frac{\lim_{x \rightarrow c^+} \mathbb{E}[Y_i X_i=x] - \lim_{x \rightarrow c^-} \mathbb{E}[Y_i X_i=x]}{\lim_{x \rightarrow c^+} \mathbb{E}[D_i X_i=x] - \lim_{x \rightarrow c^-} \mathbb{E}[D_i X_i=x]} \\ &= \dots = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 X_i=c]}_{\text{LATE at the cutoff}} \end{aligned}$	$\hat{\beta}_{\text{2SLS}}$ of the 2-stage regression $D_i = \delta + \gamma Z_i + f(X_i) + u_i$ $Y_i = \alpha + \beta \hat{D}_i + f(X_i) + e_i$ w. choice of $f(\cdot)$. Consistent, biased.	<i>Combination of the strengths and weaknesses of IV and Sharp RD</i>
DiD	(A1) same counterfactual trends across groups (A2) stable group compositions over time	$\begin{aligned} \beta_{\text{DiD}} &:= (\bar{Y}_{G_1 P_1} - \bar{Y}_{G_1 P_0}) - (\bar{Y}_{G_0 P_1} - \bar{Y}_{G_0 P_0}) \\ &= \dots = \underbrace{\mathbb{E}[Y_{i1}^1 - Y_{i1}^0 G_i=1]}_{\text{ATET}} \end{aligned}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_{it} = \beta G_i P_t + \lambda_G + \lambda_P + e_{it}$ Consistent.	👉 rules out unobserved time-invariant confounders
DiDiD	(A1) same counterfactual trends across subgroups (A2) stable subgroup compositions over time	$\begin{aligned} \beta_{\text{DiDiD}} &:= \left[(\bar{Y}_{G_1 S_1 P_1} - \bar{Y}_{G_1 S_1 P_0}) - (\bar{Y}_{G_1 S_0 P_1} - \bar{Y}_{G_1 S_0 P_0}) \right] \\ &\quad - \left[(\bar{Y}_{G_0 S_1 P_1} - \bar{Y}_{G_0 S_1 P_0}) - (\bar{Y}_{G_0 S_0 P_1} - \bar{Y}_{G_0 S_0 P_0}) \right] \\ &= \dots = \underbrace{\mathbb{E}[Y_{i1}^1 - Y_{i1}^0 G_i=1, S_i=1]}_{\text{ATET}} \end{aligned}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_{it} = \beta G_i S_i P_t + \lambda_{GS} + \lambda_{GP} + \lambda_{PS} + e_{it}$ Consistent.	👉 differences out more potential confounders than in DiD 👉 requires more variation
Event-study	(A1) random event timing (A2) stable sample composition over time	$\begin{aligned} \beta_{\text{ES}} &:= \mathbb{E}[Y_{it} t=\tau] - \mathbb{E}[Y_{it} t=\tau-1] \\ &= \dots = \underbrace{\mathbb{E}[Y_{i\tau}^1 - Y_{i\tau}^0]}_{\text{ATET}} \end{aligned}$	$\hat{\beta}_{\text{OLS}}$ of the regression $Y_{it} = \beta \mathbb{1}\{\tau\} + \sum_{t \neq \{\tau-1, \tau\}} [\beta_t \mathbb{1}\{t\}] + e_{it}$ Consistent.	👉 flexible 👉 difficult to rule out unobserved confounders

4 Analysis stage: steps for stronger causal inferences

4.1 Identification strategies only provide so much

Recall the core motivation for identification strategies:

We look for identification strategies that suggest that an independence assumption holds, as it enables us to recover an *unbiased* estimator of an average treatment effect:

- if IA, the regression of Y on D gives an unbiased estimator of the ATET (e.g., in an RCT);
- if $\cancel{\text{IA}}$ CIA + we know the correct functional form $f()$ w.r.t. the confounders X , the regression of Y on D and $f(X)$ gives an unbiased estimator of the ATET;
- in either case, if we instrument D by a valid Z , IV regression gives an unbiased estimator of a LATE.

All that identification strategies buy us is the above. This remains limited, in at least 3 important ways:

1. In observational studies, estimating causal effects relies on (i) the treatment assignment mechanism and (ii) the conditional distribution of $Y|D, X$ or *response surface*. With the $\cancel{\text{IA}}$ CIA, i.e., if ignorability holds, then we don't need to model the treatment assignment process, unbiased inferences rely only on correct modeling of the response surfaces $\mathbb{E}[Y|D=1, X]$ and $\mathbb{E}[Y|D=0, X]$; in other words, on correctly specifying the functional form w.r.t. (D, X) .

The problem is that we don't ever know this $f()$ for sure (especially if Y is not linearly related to X , if the distribution of X is quite different across treatment groups, if there are many X s...) ⁴¹, so we don't want to have to rely on $f()$... ⁴² We can see three ways to address this estimation problem:

- Focusing on appropriately controlling for the treatment assignment mechanism (e.g., with matching or weighting approaches): if the treatment assignment mechanism is properly specified we don't have to model the response surface at all (just as in a randomized experiment). This means striving to avoid overlap imbalance in our data (there, we are forced to rely on model specification instead of direct support from the data, so inferences would be vulnerable to model misspecification).
 - Focusing on correctly specifying the response surface: then we don't have to worry about correctly specifying the assignment mechanism.
 - Modeling both the treatment assignment mechanism and the response surface, to increase robustness (and potentially efficiency). This corresponds to the literature on *doubly-robust* semi-parametric estimators (Hill, 2011, section 2). These methods use estimates of the conditional outcome means as well as estimates of the propensity score, and are found to provide consistency as long as either the treatment assignment mechanism or the response surface is correctly specified. See Imbens and Wooldridge (2009, sec. 5.6).
2. An unbiased estimator $\hat{\theta}$ means that its distribution $f_{\hat{\theta}}$ (over possible trials for the given sample size) is correctly centered around the true value of the estimand θ . This does not guarantee that its realization for any particular study is close to that center value — especially with a small sample size. We might therefore want to:
 - (a) adjust as much as possible for potential imbalance between the groups, using pre-treatment data;
 - (b) consider another property: efficiency (i.e., reduce the width of the estimator's distribution).
 3. What knowledge is generated from an estimate of the ATET? Reduced forms are generally — this document is no exception — motivated by having set the RCT as gold standard. In an RCT, the treatment variable represents an intervention, and the average effect of that intervention might very well be the knowledge desired. However, in other contexts, estimating the magnitude of an effect

⁴¹Recall Angrist and Pischke (2008)'s argument that linear regression is always legitimate as it $(X\beta)$ provides the best linear approximation to the CEF $E[Y_i|D_i, X_i]$. However, if the CEF is very nonlinear, then this approximation could be terrible! Actually, even if local linearity of the regression function is a reasonable approximation, unless the linear approximation is *globally* accurate, regression may lead to severe biases (Imbens and Wooldridge, 2009).

⁴²One way to avoid possible misspecification would be to saturate the model, i.e., discretize each variable in X using indicator variables, and include a separate parameter for every possible combination of values of this set of regressors. This is rarely tractable in practice, notably with continuous X .

without identifying its underlying mechanisms might be considerably less informative (e.g., the impact of climate extremes on social instability).

This section suggests what can be done at the analysis stage (i.e., post-design, given a fixed dataset) to try to address these limitations, and generate more insightful inferences. Specifically:

1. pre-estimation: restructuring the data to improve overlap and balance w.r.t. confounders;
2. at the estimation stage: including the right covariates, and allowing for TE heterogeneity;
3. post-estimation: checking assumptions and considering external validity.

4.2 *Pre-estimation: Restructuring*

Causal inference requires the units in the treatment group to be comparable to those in the control group w.r.t. confounders X . There are two forms of departures from comparability:

- Incomplete overlap: the *support* of the distribution of X differs across the groups. Some observations have no empirical counterfactuals.
 $\rightarrow$ In these zones of no overlap, the model is forced to extrapolate, and inferences are based entirely on modeling assumptions instead of data.
- Imbalance: the *shape* of the distribution of X differs across the groups (e.g., different means, same mean but different skewness).
 $\rightarrow$ The simple difference of group averages is not, in general, a reliable estimate of the ATET.

Restructuring to balance the observed confounders The less the treatment and control groups have overlap and balance w.r.t. confounders X , the more inferences rely on the model instead of on data, and so aren't robust to model misspecification. On the contrary, if the distributions of X are similar across the groups, then, even if we misspecify the form of the relationship, we should still get a reasonable estimate of the TE (Gelman et al., 2020). To alleviate this concern of needing to specify the model correctly, we can *restructure* our sample prior to analysis, namely match groups to exhibit balance and overlap w.r.t. the confounders X (i.e., make the sample resemble one from a randomized trial: $D \perp\!\!\!\perp X$). As the estimand of interest is the ATE on the treated, we want our full sample to be representative of the *treatment* group. So we keep our treatment group intact, and restructure the control group to look like the treatment group.

$\triangle$ *Matching provides more overlap and balance, not identification. For matching to be able to capture by itself a causal effect, all the difference between the groups would need to be captured by observed X . This assumption of selection on observables is very strong, and not testable. Therefore matching is not an alternative to a design-based method.⁴³ We need exogenous variation to believe the CIA. Matching is an adjustment strategy, not an identification strategy.*

With⁴⁴ $\left\{ \begin{array}{l} (i) \text{ CIA} \\ (ii) \text{ balance \& overlap w.r.t. } X \end{array} \right. \implies \text{the difference in } \bar{Y} \text{ is an unbiased estimator of the ATET.}$

Within the context of an adjustment strategy, one can nonetheless use matching in two different ways:

• In place of regression: matching as estimation method

The 'regression with controls' estimand and the non-parametric 'covariate-matching' estimand are two different ways of balancing the X (Angrist and Pischke, 2008). In practice, the matching estimator is obtained by making comparisons for cells with the same X values, computing the difference in their Y , and averaging these differences in some way.

However, while neither the regression nor the matching *estimands* give any weight to covariate cells that don't have both treated and control observations (i.e., both estimands impose common support), the regression and matching *estimators* use modeling assumptions that implicitly involve extrapolation across cells, so cells without both treated and control observations can end up contributing to the estimates by extrapolation. Using matching as an estimation method therefore does not resolve the

⁴³Methods in which a feature in the setting approximates a randomized experiment, and we fit a model that adjusts for potential confounders: RDs, IVs... (the methods described in the previous section).

⁴⁴In other words, identification strategies and econometrics (matching, regression) are complements in the production of causal estimates. Some independence assumption is needed to give estimates (whether matching estimates or regression coefficients) a causal interpretation.

concerns with lack of overlap. In addition, estimating the standard errors of matching estimates isn't straightforward.

- **On top of regression: matching as preprocessing method**

2-step process: do matching to get comparable groups, and then do regression for further adjustment and for modeling interactions. Matching as a nonparametric preprocessing procedure is used to restructure the original sample before statistical analysis, to reduce reliance on the parametric assumptions of the subsequent regression model (Gelman et al., 2020; Ho et al., 2007).

Common distance metrics One can match units rather easily with one continuous confounder X (choosing for each treated unit the control unit with the closest value of X), or even one binary X_1 and one continuous X_2 (e.g., stratifying within subgroups defined by X_1 and then matching on X_2 within each subgroup). But it quickly gets complicated with more confounding covariates. One alternative is to define a univariate distance metric as a function of the multi-dimensional X , and match each treated unit to its nearest control unit:

- **Mahalanobis distance**

We define a distance metric that can include multiple dimensions of *closeness* between observations: $d_{ij} := \sqrt{(X_i - X_j)' \Sigma_X^{-1} (X_i - X_j)}$, where Σ_X is the sample covariance matrix. This distance metric is scale-invariant and accounts for the correlation structure of X .

- **Propensity score**

We can reduce the dimensionality to 1 by computing a unit's predicted probability of getting treated or *propensity score* $\hat{p}_i := \Pr(D_i = 1 | X_i)$ from X , and use as distance metric $d_{ij} := |\hat{p}_i - \hat{p}_j|$.

The appeal of the propensity score $p(X)$ is that if the X included in the propensity score model are sufficient to satisfy ignorability, then $p(X)$ is also sufficient to satisfy ignorability. I.e., appropriate conditioning on $p(X)$ (for instance with matching or weighting by functions of it) is sufficient to recover unbiased estimator of a TE. If $Y^1, Y^0 \perp\!\!\!\perp X$, then $Y^1, Y^0 \perp\!\!\!\perp p(X)$.

Add Rosenbaum and Rubin (1983): propensity score approach & theorem: effectively adjusts for all bias due to observable confounders.

Algorithm for propensity score matching:

1. Propensity score model: fit a logistic regression of D_i on $\{X_i\}_i$, and predict $\hat{p}_i := \Pr(D_i = 1|X_i)$.
2. Match each treated unit to its nearest control unit(s) using $\hat{p}_i$. Choose a matching algorithm: with/without replacement;^a coarse (stratifying the sample into quintile blocks of $\hat{p}_i$, e.g., 20%, and matching within block by comparing means of treatment observations $\bar{X}_1$ to control observations $\bar{X}_0$)...
3. Diagnose: assess overlap and balance. If balance is inadequate, redo steps 1-3, trying a less parsimonious model (add interactions, higher order terms of covariates...) or a less coarse matching algorithm.
 - Overlap: plot overlapping histograms w.r.t. the estimated propensity score.
 - Balance: compare the distribution of each X across the groups, before vs after matching.
4. Estimate the ATET using the restructured data. As aforementioned, one can elect to either:
 - estimate the ATET as a weighted difference in means (i.e., compute a matching estimator). *Example: Almond et al. (2005) presents both an OLS and a matching estimator.*
 - fit a regression model
 - * on D , $\hat{p}$ and $D \times \hat{p}$. See Doug Almond's "Algorithm for Estimating the Propensity Score" pdf (from Ken Chay's 2001 UC Berkeley econometrics class)
 - * on D and X using the restructured data. Gelman et al. (2020, ch. 20): *"This gives us an additional chance to adjust for differences in distributions of X that typically remain between the groups (to decrease bias and increase efficiency). The overlap and balance created by the matching should make this model more robust to potential model misspecification; that is, even if this model isn't quite right (for example, excluding a key interaction, or assuming linearity when the underlying relation is strongly nonlinear) our coefficient estimate should still be close to correct, conditional on ignorability being satisfied."*^b

^aMatching with vs without replacement is akin to a bias-variance trade-off: matching with replacement should yield better matches on average, therefore better balance and less biased TE estimators; however, it can result in over-using certain units or ignoring other close matches, i.e., missing out on important information in the data, and potentially increase the variance of the estimates.

^b Δ The standard errors from this regression are not technically correct, as: (1) matching induces correlation among the matched observations — the regression model, however, if correctly specified, should account for this by including the variables used to match; (2) the fact that the propensity score has been estimated from the data is not reflected in the calculations (Gelman et al., 2020, p. 404).

4.3 Estimation: Regression controls and TE heterogeneity

Required/forbidden regression controls (for bias) For our TE estimator to be unbiased, the identification strategy commands us to:

- ✓ adjust for *all* confounding, or more precisely, block *all* back-door paths, by adjusting for one variable along each path.⁴⁵

Δ More controls is not always better! Controlling for some confounding paths but not *all* can sometimes make the situation worse by amplifying the bias from the omitted confounder.

Middleton et al. (2016) demonstrates *bias amplification* in the simple case of two confounders, which can easily be extended. Suppose the true model is $Y_i = \alpha + \beta D_i + \delta A_i + \gamma B_i + e_i$, where A and B are confounders, but only A is observed, and $A \perp\!\!\!\perp B$ for simplicity. Should we include A ?

- Omitting A and B , i.e., regressing Y only on D , yields the bias: $\hat{\beta}_{Y|D} - \beta = \delta(D'D)^{-1}D'A + \gamma(D'D)^{-1}D'B := b_A + b_B$
 - Omitting B and including A yields the bias: $\hat{\beta}_{Y|D,A} - \beta = \frac{1}{1-R_{D|A}^2}b_B$ (where $R_{D|A}^2$ is the coefficient of determination in the regression of D on A , i.e., the amount of variation in D explained by A).
- When B is omitted, including A will actually increase net bias *iff* $|b_A + b_B| > \frac{1}{1-R_{D|A}^2}|b_B|$

Instruments are the canonical example of pure bias amplifiers: as $\delta = 0$, $b_A = 0$, so conditioning on them can only hurt. Middleton et al. (2016) further shows that group *fixed effects* can be pure bias amplifiers, even though they do not act as instruments and they absorb heterogeneity in (and

⁴⁵Technically, *reverse causality paths* $D \leftarrow \dots \leftarrow Y$ are also back-door paths, so they should be blocked too. For simplicity, we consider that there are none, the only back-door paths are therefore confounding paths.

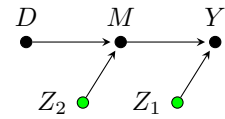
are causally related to) Y . I.e., while they are commonly thought as a harmless⁴⁶ way to account for all group-level confounding, FEs can be bias increasing *in some contexts* — e.g., when the group-level structure in Y does not covary with the group-level structure in D .

- ✗ not adjust for *post-treatment* variables that may be affected by the treatment (*intermediate outcomes*) and that are also correlated with Y (whether determined by Y —colliders—or determining Y —mediators). This would induce bias in our estimate of the total causal effect.⁴⁷

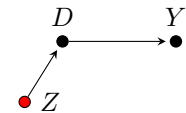
Optional good/bad regression controls (for efficiency) Separately from the identification strategy, which other covariates should we adjust for?

As a general rule of thumb, among variables which do not interfere with identification, adjusting for determinants of Y may increase the efficiency of the estimator $\hat{\beta}$, whereas determinants of D may reduce it.⁴⁸ Cinelli et al. (2021) provides a detailed and nuanced description of each case, with DAGs. Generally:

- 👉 Adjusting for *pre-treatment* covariates that have a strong association with Y (whether a direct association like Z_1 , or a mediated association like Z_2) can reduce the residual variance (the unexplained variation in Y). This will lower the standard error of $\hat{\beta}$, even though these covariates are uncorrelated with D . This applies also to data from a completely randomized experiment.



- 👈 Adjusting for determinants of D will instead reduce the variation of D and so may reduce the precision of $\hat{\beta}$ in finite samples, i.e., increase its asymptotic variance.



Given all this mess, should we even consider controlling at all, wouldn't regressing only on D be safer? Rubin (1974) states: “the investigator should be prepared to consider the possible effect of other variables besides those explicit in the experiment. Often additional variables will be ones that the investigator considers relevant because they may causally affect Y ; therefore, he may want to adjust the estimate $\Delta\bar{Y}$ and significance levels of hypotheses to reflect the values of these variables in the study. [...] An investigator who refuses to consider any additional variables is in fact saying that he does not care if $\Delta\bar{Y}$ is a bad estimate of the typical causal effect of the treatment but instead is satisfied with mathematical properties [i.e., unbiasedness] of the process by which he calculated it.” Gelman et al. (2020, p.368) further nuances the dichotomy between benefits in terms of bias vs. precision: “Under a clean randomization, adjusting for pre-treatment predictors in this way does not change what we are estimating. However, if the predictor has a strong association with the outcome it can help to bring each estimate closer (on average) to the truth, and if the randomization was less than pristine, the addition of predictors to the equation may help us adjust for systematically unbalanced characteristics across groups. Thus, this strategy has the potential to adjust for both random and systematic differences between the treatment and control groups (that is, to reduce both variance and bias), as long as these differences are characterized by differences in the pre-test.”

TE heterogeneity We expect some heterogeneity in the treatment effect β .⁴⁹ We could therefore relax the overly restrictive modeling assumption of a constant TE, and look into its variation. Options include:

- If we expect β to vary with the level of a covariate X , we can interact D with that X .⁵⁰ Gelman

⁴⁶Their harmlessness refers to bias, while they may be inefficient. Fixed effects lead to comparing much smaller changes than if we were to look at the entire range of data. If variables are measured rather imprecisely, we will be removing a lot of the signal but not any of the noise, therefore the power of the analysis goes down. See <http://www.g-feed.com/2012/12/the-good-and-bad-of-fixed-effects.html>

⁴⁷There are *sometimes* things we can learn from a regression with bad controls; see <http://www.g-feed.com/2012/10/bad-control.html> (last ¶), referring to Maccini and Yang (2009).

⁴⁸Increasing the efficiency or precision of the estimator $\hat{\beta}$ means reducing its asymptotic variance, and so its standard error. As the distribution of the estimator changes, the value of its central point — our estimate — may also change slightly.

⁴⁹For simplicity, let's assume that we are interested in *average* treatment effects, and so focus on variation in the first moment (mean) of the outcome distribution. We could also consider TE variation as variation in the second moment (variance) or in the overall outcome distribution (quantiles). Feller and Gelman (2015) provides some discussion of these cases.

⁵⁰For continuous X , consider centering it, s.t. the treatment coefficient represents the TE for individuals with the mean X score for the sample. Note that a problem when X is continuous X is that we rarely have a reason a priori to make a particular assumption of the parametric form of the interaction. E.g., we can expect that a given treatment will become increasingly effective along X , but don't know whether this relationship is linear, quadratic, exponential... Some ways to get around this

et al. (2020) recommends doing that notably with the covariates included as controls that have large estimated coefficients.

- If our data are hierarchical, i.e., there are groups $g[i]$, we can let β vary by group, and address the between-group variation by *pooling* the β_g 's in some way:
 - *Partial pooling.* The slope coefficients: 1. vary by group, 2. are themselves given a probability model: $\beta_g \sim \mathcal{F}(\mu_\beta, \sigma_\beta)$. I.e., we *model* the variation between groups. This 2nd (higher: group)-level model can have predictors or not, and has parameters of its own (the *hyperparameters* of the full model) which are also estimated from data. The group-level β_g s are partially pooled toward their group-level mean μ_β , by an amount that depends on the sample size of each group and on σ_β , which is estimated from the data.⁵¹ Partial pooling is a compromise between the arbitrary extremes of full pooling ($\sigma_\beta \rightarrow 0$) and no pooling ($\sigma_\beta \rightarrow \infty$):
 - *Full pooling.* We ignore variation between groups and impose $\beta_g = \beta$.
 - *No pooling.* We estimate independent β_g (by interacting D with group dummies), with the risk of overstating the variation between groups, i.e., overfitting the data.

△ **Don't overfit, regularize** Adding predictors and interactions increases the risk of overfitting. There are multiple ways/criteria to penalize complexity in linear regressions, regularize and select variables. Ex:

- Partial pooling (of intercepts and/or slopes)
- using priors (Bayesian inference)
- LASSO
- Elastic net regression: we minimize the sum of squared residuals plus a penalty term:

$$\{\beta_p\} = \operatorname{argmin} \text{SSR} + \lambda \sum_p [(1 - \alpha)|\beta_p| + \alpha|\beta_p|^2]$$

It overcomes the limitations of LASSO. If $\lambda = 0$, this is OLS; if $\alpha = 0$, this is LASSO.

Suggestion by Suresh Naidu: how about reporting the robustness of the estimated TE to different values of λ with LASSO — rather than arbitrary author-curated specifications across various columns of a table?

4.4 *Post-estimation:* Supporting assumptions and making predictions

4.4.1 Diagnosis tests of modeling assumptions

See section 3 of <https://clairepalandri.github.io/docs/CLRM%Estimators.pdf>.

4.4.2 Falsification tests of identifying assumptions

One can never directly *test* the identifying assumptions, i.e., prove that they hold. But one can do falsification analyzes that will either increase or decrease our confidence in them — and thus support the *internal validity* of the study. These are often referred to as *falsification* or *placebo* tests.

Show balance in $\bar{X}$ across groups Causal inference rests upon the assumption that the treatment and control groups are comparable to some extent — eventually, conditional on some covariates.

In an RCT, the identifying assumption is random assignment. If treatment was indeed randomly assigned, then the sample means of explanatory variables should be the same across the treatment and control groups (*in expectation*). RCT papers hence typically show a *balance table* of sample means of X by group.⁵²

include discretizing X (but this pushes the problem back to a specification search of a different kind: choosing cutpoints) or using flexible models such as splines.

⁵¹If slope coefficients are assumed to vary by group, it makes sense to also let intercepts vary by group (this is often already done with fixed effects), in which case a hierarchical model enables the (group-specific) varying intercepts and slopes to covary.

⁵²One should show for each X the difference of sample means between the two groups, but not a t -test of whether that difference is significantly different from zero. Indeed, a t -test tells us whether that difference could have happened by chance. The random assignment already guarantees that any difference observed would have happened by chance (unbiasedness). A t -test

Even in observational settings, it is recommended to always show a balance table, i.e., to document, for each confounder X , the difference in distributions across treatment status.

- Imbens and Wooldridge (2009, eq. 3) suggests reporting, as scale-free distance measure, the normalized difference in averages, where S_0^2 and S_1^2 are the sample variances of X in the control and treatment groups, respectively:⁵³

$$\tilde{\Delta}X := (\bar{X}_1 - \bar{X}_0) / \sqrt{S_0^2 + S_1^2}$$

- Gelman et al. (2020) suggests plotting:
 - for a binary X , the absolute difference in means $\tilde{\Delta}X := \bar{X}_1 - \bar{X}_0$. “Since the standard deviation for binary variables is determined by exclusively the mean it could be misleading to standardize.”
 - for a continuous X , the standardized difference in means, by dividing by the standard deviation of X for those observations in the inferential group (i.e., generally, in the treatment group):

$$\tilde{\Delta}X := (\bar{X}_1 - \bar{X}_0) / S_1$$

Falsification tests In observational studies, the general approach to support core identifying assumptions is to show that the specification does not find an effect when one indeed *should not* exist, e.g., by looking at an outcome which should not be affected under the identifying assumption. If the analysis picks up an effect where there isn’t one, it suggests that the identifying assumption is violated, a confounder is probably driving the relationship.⁵⁴

- **IV** The two main identifying assumptions can be tested:
 - Relevance (Z is strongly related to sorting into treatment D): directly observable in the 1st stage;
 - Exclusion restriction (Z isn’t correlated with Y through some pathway other than D). The ideal falsification test is to estimate the reduced form effect of Z on Y in a situation where Z can’t affect D . Finding an effect means Z affects Y through another channel than D , falsifying the exclusion restriction.
Ex: One can use an alternative population or an alternative outcome, that can’t be affected by the treatment but would be by potential confounders (unobserved characteristics correlated with Z and Y).
- **RD** The two main identifying assumptions can be tested:
 - Continuity or *local randomization* (all other factors determining Y evolve *smoothly* w.r.t. Z).
 - * Do other covariates jump at the cutoff c ? One can test that by estimating one’s model with Y replaced by covariates, and plotting the observations and the fitted curves. If none do, it is probable that the unobservables don’t either.
 - * There is the specific concern that units might be sorting on the running variable. If that was the case, we would expect some bunching of units at the cutoff. McCrary (2008) proposes a density test, where under H_0 , the density should be continuous at c , whereas under H_a , the density should increase at c .
 - Relevance (discontinuity in the dependence of D on Z : $D = \mathbb{1}\{Z \geq c\}$). One can test this assumption by looking at whether jumps occur at placebo cutoffs $\tilde{c}$. Imbens and Lemieux (2008) suggests taking one side of the discontinuity, using the median of the running variable in that section as placebo cutoff, and checking that there is no discontinuity in Y .
- **DiD** The two main identifying assumptions can be tested:
 - Same counterfactual trends across groups. Tests include:

in this context is therefore conceptually unsound. Hayes and Moulton (2017) explain that “the point of displaying between-arm comparisons is not to carry out a significance test, but to describe in quantitative terms how large any differences were, so that the investigator and reader can consider how much effect this may have had on the trial findings.” Angrist and Pischke (2008, table 2.2.1, p.14) seem to make that mistake in discussing the STAR experiment “The P -value in the last column is for the F -test of equality of variable means across all three groups. [...] Differences in these characteristics across the three class types are small and none are significantly different from zero. This suggests the random assignment worked as intended.”

⁵³This is different from the t -statistic for the null hypothesis of equal means: $t = (\bar{X}_1 - \bar{X}_0) / \sqrt{S_0^2/N_0 + S_1^2/N_1}$.

⁵⁴Falsification tests are different from robustness checks, which consist in estimating alternative specifications that test the same hypothesis.

- * comparing trends in the pre-period;
 - * using an alternative outcome that shouldn't be affected by the treatment, but would be affected by potential confounders, i.e., where unobservables could lead to a similar relationship, if they drove the result;
 - * using an alternative control group and checking that we find the same estimated effect;
 - * moving the event to points earlier in time, and checking that we find zero effect. (Falsely assume that the onset of treatment occurs 1, 2, 3... time periods before it actually does. Finding an effect which is statistically indistinguishable from 0 supports that the observed change is more likely due to the treatment (event) than to some alternative force.)
- Stable group composition over time. Panel data satisfies this assumption by definition; with repeated cross-sectional data, we can estimate covariate balance regressions.

• SCM

- e.g., rolling back the treatment date. moving the event to points earlier in time, and checking that we find zero effect. Ex: [Abadie et al. \(2015\)](#) does *backdating*: rewind time from the date of the treatment itself and estimate their model on an earlier (placebo) date. Finding no effect of placebo dates on output supports that any deviations found in the actual event date might be due to, e.g., structural breaks caused by the event itself.

Robustness to assumption of no unobserved confounders ([Cinelli and Hazlett, 2020](#)): new tools to evaluate: instead of arguing whether 0 confounding is plausible or not, be transparent about the amount of confounding necessary to change our conclusions. Compute sensitivity measures of point estimates and t-values. On top of a point estimate and an SE (or CI), report the Robustness Value (RV). R package: `sensemakr::` sensitivity analysis tools for OLS

Examples

DiD [Linden and Rockoff \(2008\)](#) estimates individuals' valuation of crime risk, using a hedonic pricing model. Y_{it} = property selling price, D_{it} = a registered sex offender moves in within 0.1 mile ($G_i := \mathbb{1}\{i \in 0.1 \text{ mile radius}\}$ and $P_t := \mathbb{1}\{t \in \text{post sex offender moving in}\} = \mathbb{1}\{t \geq \tau\}$). If the *same counterfactual trends* assumption didn't hold, i.e., if the prices of houses in offender areas were trending over time differently than the other houses in their neighborhood, the authors would estimate a spurious impact of the offender's arrival. They run falsification tests where they estimate the model using false arrival dates (2-3 years prior to an offender's actual arrival), and find no effect.

4.4.3 Mechanisms and external validity

Validity of a statistical analysis

- **Internal validity** = the extent to which the causal effect *in the population being studied* is *properly identified*. It is determined by how well the study can rule out alternative explanations for its findings. Threats to internal validity include omitted variables, functional form misspecification, measurement error, sample selection, heteroskedasticity and/or correlated error terms...
- **External validity** = the extent to which the inferences can be generalized to other populations and settings.
 - △ Even in randomized trials, the experimental sample often differs from the population of interest. If participation decisions are explained by observed variables, such differences can be overcome by reweighting, but participation may depend on unobserved variables.

5 Presentation

5.1 Characterizing the empirical strategy

The empirical strategy for any econometric analysis aiming for causal inference should contain — to some degree, explicitly — the following items:

1. Research question — *What causal effect of interest are we trying to estimate?*
2. Ideal experiment — *What ideal experiment would capture the causal effect?*
Define that by defining the corresponding counterfactual question first: *What would happen if certain things were changed while others were held fixed?*
3. Identification strategy — *How are the observational data at hand used to make comparisons that approximate such an experiment? Specifying notably: the identifying assumptions, what makes them satisfied, the specific effect estimated (ATET, LATE...).*
4. Estimation method (incl. assumptions made when constructing standard errors).
5. Falsification tests that bring confidence in the identifying assumptions.

All these items can be characterized before opening the dataset.

5.2 Putting the paper in perspective

In addition to the paper's empirical strategy, one may want to discuss:

- Contributions to the literature on the topic or research question
- Methodological contributions
- Internal validity of the statistical analysis
Are the identifying assumptions plausible (are there stories under which the assumptions would not hold?) Could there be measurement error? Are there unexplained results?
- External validity of the statistical analysis
 - w.r.t. policy: is there a gap between policy questions and the analyses performed?
 - w.r.t. the literature: how does the paper account for its results compared to other results in the literature?
 - w.r.t. other settings: are the results generalizable to other populations and settings?

6 Other branches of causal modeling

6.1 Which uncertainty matters? Randomization inference (RI)

“In randomization-based inference, uncertainty arises naturally from the random assignment of the treatments, rather than from the hypothesized sampling from a large population.” (Atthey and Imbens, 2017)

The frequentist inference techniques we commonly use in regression analysis correspond to *sampling-based* inference. They consider variation in sampling: the uncertainty about population parameters is induced by random sampling from the population. These methods ask: *What would have occurred under a different random sample than the one sampled?*

In causal inference studies, there is also another type of variation at play: variation in *assignment of treatment*, i.e., *design-based* uncertainty corresponding to what the regression outcome would have been under alternative randomizations of treatment assignment. In *Randomization Inference*, introduced by Fisher (1935), the basis for inference is the distribution induced by the randomization of the treatment allocation. One takes *“a design-based perspective where the properties of the estimators arises from the stochastic nature of the treatment assignment, rather than a sampling-based or model-based perspective where these properties arise from the random⁵⁵ sampling of units from a large population in combination with assumptions on this population distribution” (Atthey and Imbens, 2022)*. One asks: *What would have occurred under a different random assignment of treatment among units than the assignment observed?*

Application to hypothesis testing Both sampling-based and design-based inference follow the same approach to hypothesis testing: we formulate a null hypothesis that represents a fact about the data we’ll try to refute. In causal inference, it is generally a hypothesis of no effect. We then derive a test statistic T s.t. when H_0 is true, T has a specific distribution, and we look at where the value of T for our observed data T_{obs} lies within that distribution. The furthest in the tails, the less likely these observed data were under the null hypothesis, therefore the higher the confidence against it.

In randomization inference, considering the *sharp* null hypothesis of no effect for any unit,⁵⁶ we can simply use β as the test-statistic and obtain its empirical distribution under H_0 . Indeed:

- If there is no effect for any unit, then a unit’s potential outcomes are identical: the observed outcome is also the counterfactual. Under H_0 , our data therefore represent the outcomes of all possible experiments. We know the exact value of all the missing potential outcomes.
- If we construct all possible random assignments, estimate $\hat{\beta}$ for each, the resulting distribution of $\hat{\beta}$ is therefore the reference distribution under H_0 .
- We look at where our actual $\hat{\beta}_{\text{obs}}$ falls in the reference distribution; if in the tails, e.g., such that only 2% of all random assignments produce a $\hat{\beta} \geq \hat{\beta}_{\text{obs}}$, our one-tailed p-value is 0.02.

In practice: simulation When *all* possible random assignments can be simulated, the reference distribution is known, thus RI produces *exact* p-values. In practice, the number of possible assignments is generally huge, so we don’t simulate all of them but many, to approximate the reference distribution, and compute approximate p-values. We repeat a large number of times (e.g., 10000) the following procedure:⁵⁷

1. Re-assign treatment randomly, i.e., draw from the *randomization set*⁵⁸ (respecting the structure of the original assignment mechanism, e.g., within strata), thereby generating fake treatment statuses.
2. Estimate the regression model using these fake treatments, and store the $\hat{\beta}$ s.

⁵⁵At this point the term ‘randomization’ might seem confusing, as *both* approaches assume and build inference from randomness: in the traditional approach, that of the *sample*; in the design approach, of the *treatment assignment*. There is a subtle difference: in the first the sample isn’t *randomized* but simply *random*, i.e., taken randomly, whereas in the second, because assignment is made in a random fashion, the resulting treatment is first randomized, and therefore random. RI is aptly named.

⁵⁶Note that this is substantially different from the usual null hypothesis in sampling-based inference of *no average effect*.

⁵⁷RI is a simulation approach, like Bootstrap, however Bootstrap considers variation from sampling. A Bootstrap procedure resamples observations from our actual sample (which is fair, as we assumed it was representative of the population), with replacement, to simulate how *sampling* variation would affect our results.

⁵⁸(Rubin, 1974) defines the “randomization set” as “the set of allocations that were equally likely to be observed given the randomization plan.” Ex: for a completely randomized experiment of $2N$ trials, where N is assigned to each treatment arm, there are $\binom{2N}{N}$ possible allocations.

We obtain a distribution for the $\hat{\beta}$ s.

Sampling-based inference	Randomization inference
H_0, H_a	
H_0 : No <i>average</i> effect: $\mathbb{E}[Y_i^1] - \mathbb{E}[Y_i^0] = 0$	H_0 : <i>Sharp</i> no effect: $Y_i^1 - Y_i^0 = 0, \forall i$
H_a : An average effect: $\mathbb{E}[Y_i^1] - \mathbb{E}[Y_i^0] \neq 0$	H_a : $\exists i$ s.t. $Y_i^1 - Y_i^0 \neq 0$
distribution of T under H_0	
$T := \frac{\hat{\beta} - \beta_0}{\text{SE}[\hat{\beta}]} = \frac{\hat{\beta} - 0}{\text{SE}[\hat{\beta}]}$	$T := \hat{\beta}$
Under H_0 , the distribution of T across all random samples converges (as $n \rightarrow \infty$) to a Student's $\mathcal{T}$ distribution. $\rightarrow$ We compute the parameters of this distribution. $\rightarrow$ The <i>asymptotic</i> distribution of T (across all random samples) = the <i>sampling distribution</i> under H_0 .	Under H_0 , how the treatment was randomly assigned wouldn't change the observed outcomes; but it would change the value of T . $\rightarrow$ We compute T for all possible random assignments. $\rightarrow$ The <i>exact</i> distribution of T (across all random assignments) = the <i>reference distribution</i> under H_0 .
2-tailed p-value = $\Pr [\text{observing a } T > T_{\text{obs}}]$ under H_0	
= share of the distribution that is $> T_{\text{obs}}$	
= $\Pr(\text{the observed difference between groups would have been observed})$ if they had been drawn from underlying sampling frames with no mean difference.	= $\Pr(\text{the observed difference between groups would have been observed})$ if the treatment effect were in fact 0 for every subject.
$\implies$ Given e.g. a rejection threshold $\alpha = 0.05$, the test will erroneously reject H_0 less than 5% of the time	

Why choose randomization-based inference instead of sampling-based inference?

- Conceptually, there is sometimes no true sampling variation to speak of. Suppose we observed the universe of y outcomes, then there is no sampling from a large population, making sampling-based p -values meaningless, $\text{SE} = 0$.⁵⁹ Regardless, the core uncertainty within a causal study is not solely driven by the universe of possible samples, but also by the universe of possible treatment assignments.
- RI is not confined to large samples. As we don't have to appeal to the asymptotic properties of an estimator, it allows us to make inferences about causal effects even in settings where assuming an infinite number of treatment units may not be credible.
- RI is not confined to normally distributed outcomes. The method can be applied to all sorts of outcomes, such as counts, durations, ranks (Gerber and Green, 2012, p.63).
- RI salvages inference with particular clustered designs
 - *Small number of assignment clusters*: When the number of clusters is small, cluster-robust standard errors are downwardly biased. RI circumvents this problem as the reference distribution is calculated based on the set of possible clustered assignments, which takes into account the sampling variability associated with clustered assignment.
 - *Assignment clusters without well-defined boundaries*: if the assignment clustering isn't within well-defined boundaries, one can't rely on common methods to estimate correct standard errors (clusters can't be defined; other sandwich-type covariance matrix estimators require additional modeling assumptions...). Ex: weather variables such as rainfall are often used as a strategy for causal inference, as rainfall shocks are as-if randomly assigned. However, the assignment of rainfall is highly correlated across space in an unformalizable structure. Cooperman (2017) uses national draws of historical rainfall patterns as potential randomizations, allowing her to preserve patterns of spatial dependence while remaining agnostic about the specific form of the clustering.⁶⁰
- Apparently RI is somewhat more robust to the presence of leverage in a few observations. Young (2019) collected over fifty experimental (lab and field) articles from the American Economic Review, American Economic Journal: Applied, and American Economic Journal: Economic Policy. He then reanalyzed these papers, using the authors' models, by dropping one observation or cluster and

⁵⁹While it is indeed possible to observe the value of a variable for all the units in a population, one rarely observes all the possible range of values that units could have taken. Thinking of that universe of values as the relevant population alleviates the conceptual concern.

⁶⁰Note that the use of historical data is disputable if climate change changes the distribution across years.

reestimating the entire model, repeatedly. He found that with the removal of just one observation, 35% of 0.01-significant reported results in the average paper can be rendered insignificant at that level, 16% of 0.01-insignificant reported results can be found to be significant at that level. In the typical paper, randomization inference found individual treatment effects that were 13 to 22 percent fewer significant results than what the authors' own analysis had discovered.

Limitations

- RI does not accommodate null hypotheses such as the average effect of the treatment is zero, against the alternative hypothesis of a non zero average effect.
- RI is a method for hypothesis testing — not for constructing confidence intervals!
 - ⚠ The *reference distribution under H_0* does not give confidence intervals for $\hat{\beta}$. It is instead the set of all possible estimated values of $\hat{\beta}$ when the true $\beta = 0$. This does not represent our statistical uncertainty about $\hat{\beta}$, it only enables us to compute p-values for the sharp null hypothesis of no effect.
 - ↔ It is legitimate to ask whether RI is even worth it then, given the heavy criticism of the 2-way binary approach to statistical hypothesis testing, based on the NHST falsificationist paradigm and the formulation of binary statements of ‘statistical significance’ from a p-value threshold.
- RI may also be used for construction of confidence intervals, but this requires additional assumptions.
 - Rosenbaum (2002, p.45) proposes a method by *inverting* the hypothesis test.
 - Gerber and Green (2012, p.67) proposes a simpler — but less accurate — method, and argues that the two methods tend to produce similar results, especially in large samples.
 - Barrios et al. (2012, eq. (4.2)) gives the exact conditional (randomization-based) variance of $\hat{\beta}$ (in the univariate linear regression model of Y on D) under the assumption of a homogeneous treatment effect, based on Neyman (1923) (unfortunately, the proof is omitted):

$$\mathbb{V}[\hat{\beta}_{\text{OLS}}|e] = \frac{N}{N_0 N_1 (N - 2)} \sum_i (e_i - \bar{e})^2, \quad \text{where } N_1 := \sum_i D_i, \quad N_0 := N - N_1$$

6.2 Structural Equation Models (SEMs)

Structural Equation Models are probabilistic models that unite multiple predictor and response variables in a single causal network.

SEMs are increasingly popular in ecological research. They are often represented using path diagrams, a.k.a. directed acyclic graphs (DAG), where arrows indicate directional relationships between observed variables.

Implicit assumptions — what separate SEMs from traditional modeling approaches:

1. SEMs implicitly assume that the relationships among variables (paths) are causal. This is a big leap from the traditional statistics' "correlation does not imply causation." By using pre-existing knowledge of the system, one makes an informed hypothesis about the causal structure of the variables, and the SEM explicitly tests this supposed causal structure.
2. Variables can be both predictors and responses. A SEM is thereby useful for testing and quantifying indirect (cascading) effects that would otherwise go unrecognized by any single model.

Traditional SEM	Piecewise SEM
Estimation: Coefficients are estimated simultaneously in a single variance-covariance matrix of all variables; typically by MLE. Goodness-of-fit: = discrepancy between the observed and predicted covariance matrices. χ^2 test: the χ^2 statistic describes the agreement between the 2 matrices. Assumptions • Independent errors (no underlying structure) • Normal errors Limits • <i>Assumptions often violated in ecological research: e not independent (spatial or temporal correlation in observational studies), distribution not normal (count data $\sim$ Poisson)...</i> • computationally intensive (depending on the sizes of the variance-covariance matrix); • if variables are nested, then the sample size is limited to the use of variables at the highest level of the hierarchy. Can shrink our sample and reduce the power of the analysis...	Estimation: Decompose the network and estimate each relationship separately (estimate m separate vcov matrices). Then piece the m paths together for inferences about the entire SEM. ⇒ Much easier to estimate than a single vcov matrix → can estimate large networks ⇒ Flexible: can incorporate many model structures, distributions... using extensions of linear reg (random effects, hierarchical models, non-normal responses, spatial correlation...) Goodness-of-fit: No formal χ^2 test. Instead: <i>tests of directed separation</i>: are any paths missing from the model? The <i>basis set</i> := all k pair relationships unspecified in the model (i.e., independence claims). Test whether are indeed not significant (controlling for variables on which these paths are conditional), keep the p-value. From the k p-values, calculate Fisher's C statistic $C = -2 \sum_{i=1}^k \ln(p_i) \sim \chi^2(2k)$. If C's p-value > 0.05, accept the model. This approach is vulnerable to model misspecification. <i>Rmk: we can compute an AIC score for the SEM, for model comparisons: $AIC = C + 2k \frac{n}{n-k-1}$</i>

6.3 Structural Vector Autoregression (SVAR)

Add [Ghanem and Smith \(2021\)](#), and [Greene 2002](#), section 19.6.7.

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A Maths of potential outcomes

The steps that were overlooked in the main document are provided here in blue.

2.1.1 The original selection bias problem

$$\begin{aligned}
 \mathbb{E}[Y_i|D_i=1] - \mathbb{E}[Y_i|D_i=0] &= \mathbb{E}[Y_i^1|D_i=1] - \mathbb{E}[Y_i^0|D_i=0] \quad (\text{definition of potential outcomes}) \\
 &= \mathbb{E}[Y_i^1|D_i=1] - \mathbb{E}[Y_i^0|D_i=1] + \mathbb{E}[Y_i^0|D_i=1] - \mathbb{E}[Y_i^0|D_i=0] \\
 &= \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 | D_i=1]}_{\text{ATET}} + \underbrace{\mathbb{E}[Y_i^0 | D_i=1] - \mathbb{E}[Y_i^0 | D_i=0]}_{\text{selection bias}}
 \end{aligned}$$

The same demonstration holds conditional on X_i , i.e., within each stratum of X_i :

$$\begin{aligned}
 \mathbb{E}[Y_i|D_i=1, X_i] - \mathbb{E}[Y_i|D_i=0, X_i] &= \mathbb{E}[Y_i^1|D_i=1, X_i] - \mathbb{E}[Y_i^0|D_i=0, X_i] \\
 &= \mathbb{E}[Y_i^1|D_i=1, X_i] - \mathbb{E}[Y_i^0|D_i=1, X_i] + \mathbb{E}[Y_i^0|D_i=1, X_i] - \mathbb{E}[Y_i^0|D_i=0, X_i] \\
 &= \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 | D_i=1, X_i]}_{\text{ATET for given } X_i} + \underbrace{\mathbb{E}[Y_i^0 | D_i=1, X_i] - \mathbb{E}[Y_i^0 | D_i=0, X_i]}_{\text{selection bias for given } X_i}
 \end{aligned}$$

2.1.2 Average treatment effects as regression slopes and endogeneity

Simplest setting: binary D , no X

The population OLS slope estimand simplifies to the difference in average observed outcomes, which we show itself simplifies to a condition on the *causal disturbance* u_i^c :

$$\begin{aligned}
 \beta_{\text{OLS}} &:= \frac{\text{cov}[Y_i, D_i]}{\mathbb{V}[D_i]} = \frac{\mathbb{E}[Y_i D_i] - \mathbb{E}[Y_i]\mathbb{E}[D_i]}{\mathbb{E}[D_i^2] - \mathbb{E}[D_i]^2} \\
 &= \frac{\mathbb{E}[Y_i|D_i=1] \Pr(D_i=1) - \left(\mathbb{E}[Y_i|D_i=0] \Pr(D_i=0) + \mathbb{E}[Y_i|D_i=1] \Pr(D_i=1) \right) \Pr(D_i=1)}{\Pr(D_i=1) - \Pr(D_i=1)^2} \\
 &= \frac{\mathbb{E}[Y_i|D_i=1] \Pr(D_i=1) (1 - \Pr(D_i=1)) - \mathbb{E}[Y_i|D_i=0] \Pr(D_i=0) \Pr(D_i=1)}{\Pr(D_i=1)(1 - \Pr(D_i=1))} \\
 &= \frac{\mathbb{E}[Y_i|D_i=1] \Pr(D_i=1) \Pr(D_i=0) - \mathbb{E}[Y_i|D_i=0] \Pr(D_i=0) \Pr(D_i=1)}{\Pr(D_i=1) \Pr(D_i=0)} \\
 &= \mathbb{E}[Y_i | D_i=1] - \mathbb{E}[Y_i | D_i=0]
 \end{aligned}$$

From the causal regression-form decomposition, $Y_i = \alpha^c + \beta^c D_i + u_i^c$ (where β^c is the ATET), hence:

$$\begin{cases} \mathbb{E}[Y_i|D_i=1] = \alpha + \beta^c + \mathbb{E}[u_i^c|D_i=1] \\ \mathbb{E}[Y_i|D_i=0] = \alpha + \mathbb{E}[u_i^c|D_i=0] \end{cases} \Rightarrow \beta_{\text{OLS}} = \beta^c + \mathbb{E}[u_i^c|D_i=1] - \mathbb{E}[u_i^c|D_i=0].$$

We can also recover again the expression with selection bias, from the expression of the causal disturbance $u_i^c := Y_i^0 - \mathbb{E}[Y_i^0 | D_i=0] + (\beta_i - \mathbb{E}[\beta_i | D_i=1]) D_i$:

$$\begin{cases} \mathbb{E}[u_i^c|D_i=1] = \mathbb{E}[Y_i^0|D_i=1] - \mathbb{E}[Y_i^0|D_i=0] \\ \mathbb{E}[u_i^c|D_i=0] = 0 \end{cases} \Rightarrow \mathbb{E}[u_i^c|D_i=1] - \mathbb{E}[u_i^c|D_i=0] = \mathbb{E}[Y_i^0|D_i=1] - \mathbb{E}[Y_i^0|D_i=0]$$

Therefore $\beta_{\text{OLS}} = \dots = \mathbb{E}[Y_i | D_i=1] - \mathbb{E}[Y_i | D_i=0] = \dots = \text{ATET} + \text{selection bias}$.

With covariates $\mathbf{X}$

For simplicity, consider a single discrete X_i .

- Matching estimand

$$\begin{aligned}\beta_M &= \sum_x \delta_x \Pr(X_i=x \mid D_i=1) = \sum_x \delta_x \frac{\Pr(X_i=x, D_i=1)}{\Pr(D_i=1)} = \sum_x \delta_x \frac{\Pr(D_i=1 \mid X_i=x) \Pr(X_i=x)}{\Pr(D_i=1)} \\ &= \frac{1}{\Pr(D_i=1)} \sum_x \delta_x \Pr(D_i=1 \mid X_i=x) \Pr(X_i=x) \\ &= \frac{\sum_x \delta_x \Pr(D_i=1 \mid X_i=x) \Pr(X_i=x)}{\sum_x \Pr(D_i=1 \mid X_i=x) \Pr(X_i=x)}\end{aligned}$$

- OLS estimand

The demonstration uses the Frisch-Waugh-Lovell theorem (Angrist and Pischke, 2008, p.55).

3.1 IV

IV estimand

$$\begin{aligned}\beta_{IV} &:= \frac{\text{cov}[Y_i, Z_i]}{\text{cov}[D_i, Z_i]} = \frac{\mathbb{E}[Y_i Z_i] - \mathbb{E}[Y_i] \mathbb{E}[Z_i]}{\mathbb{E}[D_i Z_i] - \mathbb{E}[D_i] \mathbb{E}[Z_i]} \\ &= \frac{\mathbb{E}[Y_i \mid Z_i=1] \Pr(Z_i=1) - (\mathbb{E}[Y_i \mid Z_i=1] \Pr(Z_i=1) + \mathbb{E}[Y_i \mid Z_i=0] \Pr(Z_i=0)) \Pr(Z_i=1)}{\mathbb{E}[D_i \mid Z_i=1] \Pr(Z_i=1) - (\mathbb{E}[D_i \mid Z_i=1] \Pr(Z_i=1) + \mathbb{E}[D_i \mid Z_i=0] \Pr(Z_i=0)) \Pr(Z_i=1)} \\ &= \frac{\mathbb{E}[Y_i \mid Z_i=1] (1 - \Pr(Z_i=1)) - \mathbb{E}[Y_i \mid Z_i=0] \Pr(Z_i=0)}{\mathbb{E}[D_i \mid Z_i=1] (1 - \Pr(Z_i=1)) - \mathbb{E}[D_i \mid Z_i=0] \Pr(Z_i=0)} \\ &= \frac{\mathbb{E}[Y_i \mid Z_i=1] - \mathbb{E}[Y_i \mid Z_i=0]}{\mathbb{E}[D_i \mid Z_i=1] - \mathbb{E}[D_i \mid Z_i=0]}\end{aligned}$$

The identifying assumptions then reduce it to the LATE on the compliers:

- Numerator: $\mathbb{E}[Y_i \mid Z_i=1] - \mathbb{E}[Y_i \mid Z_i=0] =$

$$\begin{aligned}&= \mathbb{E}[Y_i \mid Z_i=1, D_i^0=0, D_i^1=0] P(D_i^0=0, D_i^1=0) - \mathbb{E}[Y_i \mid Z_i=0, D_i^0=0, D_i^1=0] P(D_i^0=0, D_i^1=0) \\ &\quad + \mathbb{E}[Y_i \mid Z_i=1, \text{---} 0, \text{---} 1] P(\text{---} 0, \text{---} 1) - \mathbb{E}[Y_i \mid Z_i=0, \text{---} 0, \text{---} 1] P(\text{---} 0, \text{---} 1) \\ &\quad + \mathbb{E}[Y_i \mid Z_i=1, \text{---} 1, \text{---} 0] P(\text{---} 1, \text{---} 0) - \mathbb{E}[Y_i \mid Z_i=0, \text{---} 1, \text{---} 0] P(\text{---} 1, \text{---} 0) \\ &\quad + \mathbb{E}[Y_i \mid Z_i=1, \text{---} 1, \text{---} 1] P(\text{---} 1, \text{---} 1) - \mathbb{E}[Y_i \mid Z_i=0, \text{---} 1, \text{---} 1] P(\text{---} 1, \text{---} 1) \\ &= \mathbb{E}[Y_i^0 \mid D_i^0=0, D_i^1=0] P(D_i^0=0, D_i^1=0) - \mathbb{E}[Y_i^0 \mid D_i^0=0, D_i^1=0] P(D_i^0=0, D_i^1=0) \\ &\quad + \mathbb{E}[Y_i^1 \mid \text{---} 0, \text{---} 1] P(\text{---} 0, \text{---} 1) - \mathbb{E}[Y_i^0 \mid \text{---} 0, \text{---} 1] P(\text{---} 0, \text{---} 1) \\ &\quad + \mathbb{E}[Y_i^0 \mid \text{---} 1, \text{---} 0] P(\text{---} 1, \text{---} 0) - \mathbb{E}[Y_i^1 \mid \text{---} 1, \text{---} 0] P(\text{---} 1, \text{---} 0) \\ &\quad + \mathbb{E}[Y_i^1 \mid \text{---} 1, \text{---} 1] P(\text{---} 1, \text{---} 1) - \mathbb{E}[Y_i^1 \mid \text{---} 1, \text{---} 1] P(\text{---} 1, \text{---} 1) \\ &= \mathbb{E}[Y_i^1 - Y_i^0 \mid D_i^0=0, D_i^1=1] P(D_i^0=0, D_i^1=1) - \mathbb{E}[Y_i^1 - Y_i^0 \mid D_i^0=1, D_i^1=0] P(D_i^0=1, D_i^1=0) \\ &= \mathbb{E}[Y_i^1 - Y_i^0 \mid D_i^0=0, D_i^1=1] P(D_i^0=0, D_i^1=1) \text{ as the probability of defiers is 0}\end{aligned}$$
 - Denominator:

$$\begin{aligned}\mathbb{E}[D_i \mid Z_i=1] - \mathbb{E}[D_i \mid Z_i=0] &= \mathbb{E}[D_i^1 - D_i^0] \\ &= 1 \times P(D_i^1 - D_i^0 = 1) + 0 \times P(D_i^1 - D_i^0 = 0) - 1 \times P(D_i^1 - D_i^0 = -1) \\ &= P(D_i^0=0, D_i^1=1) \text{ as the probability of defiers is 0}\end{aligned}$$
- $$\implies \frac{\mathbb{E}[Y_i \mid Z_i=1] - \mathbb{E}[Y_i \mid Z_i=0]}{\mathbb{E}[D_i \mid Z_i=1] - \mathbb{E}[D_i \mid Z_i=0]} = \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid D_i^0=0, D_i^1=1]}_{\text{LATE on the compliers}}$$

Counting and characterizing compliers to get more out of the LATE

- Size of the complier group: It is the Wald first-stage, as, given monotonicity:

$$\Pr(D_i^1 > D_i^0) = \Pr(D_i^1 - D_i^0 = 1) = \mathbb{E}[D_i^1 - D_i^0] = \mathbb{E}[D_i^1] - \mathbb{E}[D_i^0] = \mathbb{E}[D_i | Z_i=1] - \mathbb{E}[D_i | Z_i=0]$$

- Share of treated that are compliers:

$$\begin{aligned} \Pr(D_i^1 > D_i^0 \mid D_i=1) &= \frac{\Pr(D_i^1 > D_i^0, D_i=1)}{\Pr(D_i=1)} = \frac{\Pr(D_i=1 \mid D_i^1 > D_i^0) \Pr(D_i^1 > D_i^0)}{\Pr(D_i=1)} \\ &= \frac{\Pr(Z_i=1 \mid D_i^1 > D_i^0) \Pr(D_i^1 > D_i^0)}{\Pr(D_i=1)} \\ &= \frac{\Pr(Z_i=1) \Pr(D_i^1 > D_i^0)}{\Pr(D_i=1)} \\ &= \frac{\Pr(Z_i=1) \times (\mathbb{E}[D_i | Z_i=1] - \mathbb{E}[D_i | Z_i=0])}{\Pr(D_i=1)} \\ &= \frac{\Pr(\text{instrument is switched on}) \times \text{1st stage}}{\text{share treated}} \end{aligned}$$

- Distribution of covariates X for compliers:
 - For binary characteristics X, we can compute relative likelihoods:

$$\begin{aligned} \frac{\Pr(X_i=1 \mid D_i^1 > D_i^0)}{\Pr(X_i=1)} &= \frac{\Pr(X_i=1, D_i^1 > D_i^0)}{\Pr(X_i=1) \Pr(D_i^1 > D_i^0)} = \frac{\Pr(D_i^1 > D_i^0 \mid X_i=1)}{\Pr(D_i^1 > D_i^0)} \\ &= \frac{\mathbb{E}[D_i \mid Z_i=1, X_i=1] - \mathbb{E}[D_i \mid Z_i=0, X_i=1]}{\mathbb{E}[D_i \mid Z_i=1] - \mathbb{E}[D_i \mid Z_i=0]} \\ &= \frac{\text{1st stage} \mid X_i=1}{\text{1st stage}} \end{aligned}$$

3.2 RD

Sharp RD estimand

$$\begin{aligned} \beta_{\text{RD}} &:= \lim_{x \rightarrow c^+} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \rightarrow c^-} \mathbb{E}[Y_i \mid X_i = x] \\ &= \lim_{x \rightarrow c^+} \mathbb{E}[Y_i^1 \mid X_i = x] - \lim_{x \rightarrow c^-} \mathbb{E}[Y_i^0 \mid X_i = x] \\ &= \mathbb{E}[Y_i^1 \mid X_i = c] - \mathbb{E}[Y_i^0 \mid X_i = c] \\ &= \underbrace{\mathbb{E}[Y_i^1 - Y_i^0 \mid X_i = c]}_{\text{LATE at the cutoff}} \end{aligned}$$

Fuzzy RD estimand

$$\beta_{\text{IV}} := \frac{\lim_{x \rightarrow c^+} \mathbb{E}[Y_i \mid X_i = x] - \lim_{x \rightarrow c^-} \mathbb{E}[Y_i \mid X_i = x]}{\lim_{x \rightarrow c^+} \mathbb{E}[D_i \mid X_i = x] - \lim_{x \rightarrow c^-} \mathbb{E}[D_i \mid X_i = x]} = \lim_{\delta \rightarrow 0} \frac{\mathbb{E}[Y_i \mid c < X_i < c + \delta] - \mathbb{E}[Y_i \mid c - \delta < X_i < c]}{\mathbb{E}[D_i \mid c < X_i < c + \delta] - \mathbb{E}[D_i \mid c - \delta < X_i < c]}$$

$$\mathbb{E}[Y_i \mid c < X_i < c + \delta] - \mathbb{E}[Y_i \mid c - \delta < X_i < c] \simeq \gamma \beta$$

$$\mathbb{E}[D_i \mid c < X_i < c + \delta] - \mathbb{E}[D_i \mid c - \delta < X_i < c] \simeq \gamma$$

$$\text{Therefore, } \beta_{\text{IV}} = \frac{\gamma \beta}{\gamma} = \beta$$

3.3 DiD, DiDiD, Event-study

DiD estimand

$$\begin{aligned}
\beta_{\text{DiD}} &:= (\bar{Y}_{G_1 P_1} - \bar{Y}_{G_1 P_0}) - (\bar{Y}_{G_0 P_1} - \bar{Y}_{G_0 P_0}) \\
&:= (\mathbb{E}[Y_{i1} \mid G_i=1] - \mathbb{E}[Y_{i0} \mid G_i=1]) - (\mathbb{E}[Y_{i1} \mid G_i=0] - \mathbb{E}[Y_{i0} \mid G_i=0]) \\
&= (\mathbb{E}[Y_{i1}^1 \mid G_i=1] - \mathbb{E}[Y_{i0} \mid G_i=1]) - (\mathbb{E}[Y_{i1}^0 \mid G_i=0] - \mathbb{E}[Y_{i0} \mid G_i=0]) \\
&= \mathbb{E}[Y_{i1}^1 - Y_{i0} \mid G_i=1] - \mathbb{E}[Y_{i1}^0 - Y_{i0} \mid G_i=0] \\
&= \mathbb{E}[Y_{i1}^1 - Y_{i0} \mid G_i=1] - \mathbb{E}[Y_{i1}^0 - Y_{i0} \mid \mathbf{G}_i=\mathbf{1}] \quad (\text{assumption of parallel trends}) \\
&= \underbrace{\mathbb{E}[Y_{i1}^1 - Y_{i1}^0 \mid G_i=1]}_{\text{ATET}}
\end{aligned}$$

DiDiD estimand

$$\begin{aligned}
\beta_{\text{DiDiD}} &:= \left[(\bar{Y}_{G_1 S_1 P_1} - \bar{Y}_{G_1 S_1 P_0}) - (\bar{Y}_{G_1 S_0 P_1} - \bar{Y}_{G_1 S_0 P_0}) \right] - \left[(\bar{Y}_{G_0 S_1 P_1} - \bar{Y}_{G_0 S_1 P_0}) - (\bar{Y}_{G_0 S_0 P_1} - \bar{Y}_{G_0 S_0 P_0}) \right] \\
&= (\mathbb{E}[Y_{i1} - Y_{i0} \mid G_1, S_1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid G_1, S_0]) - (\mathbb{E}[Y_{i1} - Y_{i0} \mid G_0, S_1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid G_0, S_0]) \\
&= (\mathbb{E}[Y_{i1}^1 - Y_{i0} \mid G_1, S_1] - \mathbb{E}[Y_{i1}^0 - Y_{i0} \mid G_1, S_0]) - (\mathbb{E}[Y_{i1}^0 - Y_{i0} \mid G_0, S_1] - \mathbb{E}[Y_{i1}^0 - Y_{i0} \mid G_0, S_0]) \\
&= (\mathbb{E}[Y_{i1}^1 - Y_{i0} \mid G_1, S_1] - \cancel{\mathbb{E}[Y_{i1}^0 - Y_{i0} \mid G_1, S_0]}) - (\mathbb{E}[Y_{i1}^0 - Y_{i0} \mid \mathbf{G}_1, S_1] - \cancel{\mathbb{E}[Y_{i1}^0 - Y_{i0} \mid \mathbf{G}_1, S_0]}) \quad (\parallel \text{ trends}) \\
&= \mathbb{E}[Y_{i1}^1 - Y_{i0} \mid G_1, S_1] - \mathbb{E}[Y_{i1}^0 - Y_{i0} \mid G_1, S_1] \\
&= \underbrace{\mathbb{E}[Y_{i1}^1 - Y_{i1}^0 \mid G_1, S_1]}_{\text{ATET}}
\end{aligned}$$